

How lung cancer arises, based on complexity theory
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Introduction

This is my fourth paper in a series discussing the top causes of U.S. cancer death and how they develop according to complexity theory (see prior papers on [colon cancer](#), [pancreatic cancer](#) and [breast cancer](#)). It builds on and updates my [earlier paper on lung cancer](#).

This paper discusses these aspects of lung cancer:

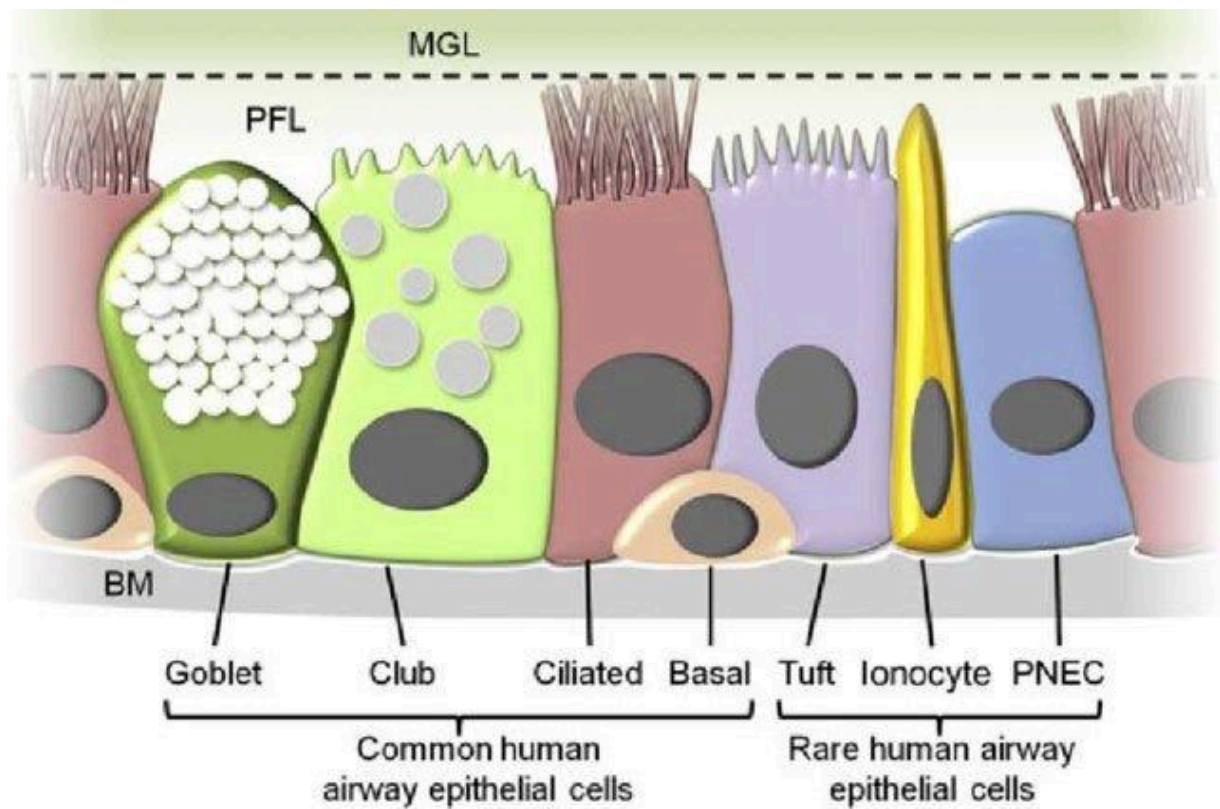
- I. Anatomy and histology
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I. Anatomy and histology of lung cancer

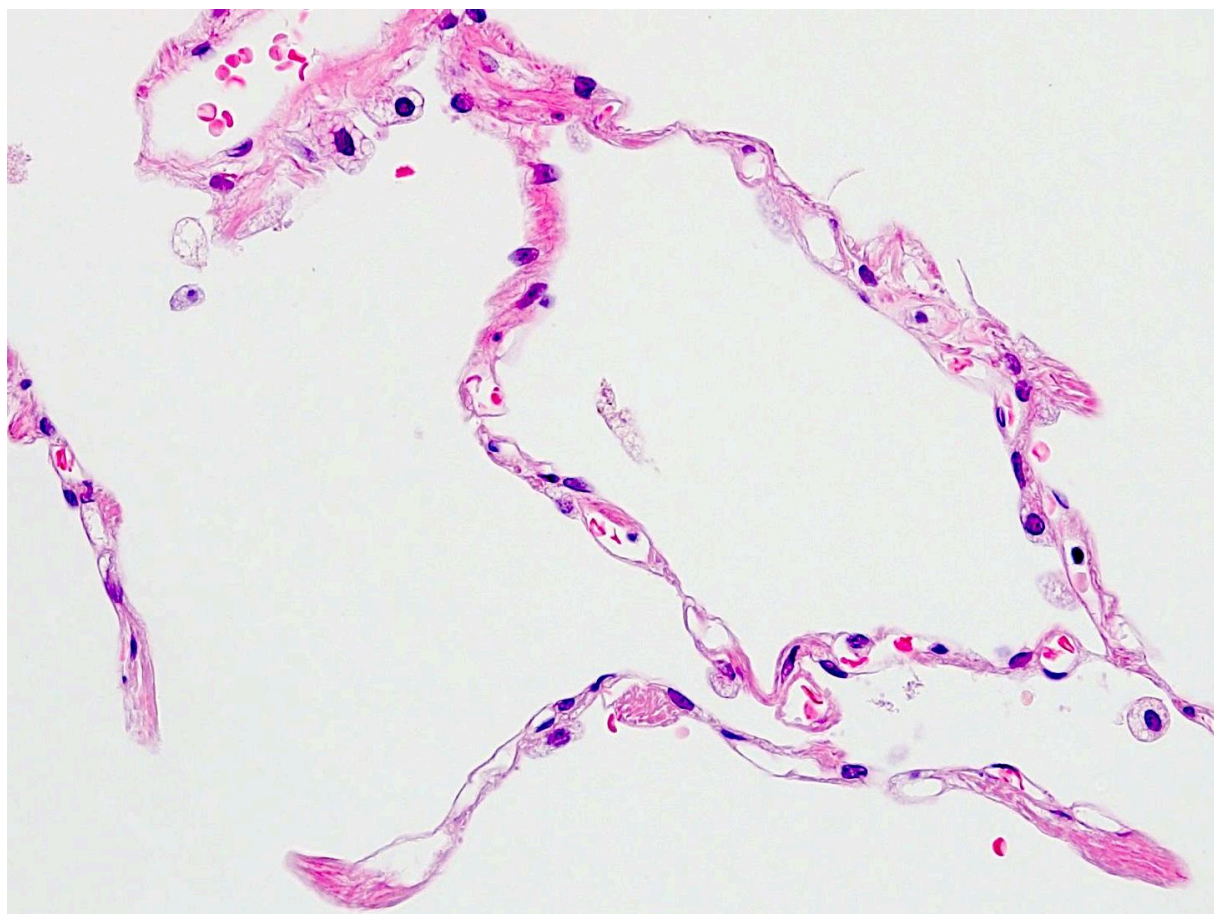
Central lung cancers arise in the main bronchi and are most often squamous cell carcinoma or small cell lung cancer. Peripheral lung cancers arise in the alveoli or bronchioles and are most often adenocarcinoma.

For the most common lung cancers, the cells of origin, anatomic locations, normal functions and associated malignancies are as follows:

Cell of origin	Typical anatomic location	Normal function	Lung cancer types commonly associated	
Alveolar type II pneumocytes	Alveoli, terminal respiratory units, lung periphery	Surfactant production; progenitor for alveolar epithelium	Adenocarcinoma; adenocarcinoma in situ; minimally invasive adenocarcinoma; lepidic, acinar, papillary, micropapillary patterns	
Club cells (Clara cells)	Terminal and respiratory bronchioles	Detoxification of inhaled toxins; secretion of protective proteins; bronchiolar progenitor cell	Adenocarcinoma, particularly peripheral tumors; some mucinous adenocarcinomas	
Basal cells	Trachea and proximal bronchi	Stem/progenitor cells for bronchial epithelium; regenerate ciliated and secretory cells	Squamous cell carcinoma; some adenosquamous carcinomas	
Pulmonary neuroendocrine cells	Scattered singly or in neuroepithelial bodies, mainly central airways	Secretion of neuropeptides and amines; regulation of airway tone and lung development	Small cell lung carcinoma; large cell neuroendocrine carcinoma; typical and atypical carcinoid tumors	
Goblet (mucous) cells	Bronchi and bronchioles	Mucin secretion for airway lubrication and host defense	Mucinous adenocarcinoma; mixed adenocarcinomas with mucinous differentiation	
Ciliated bronchial epithelial cells	Trachea and bronchi	Mucociliary clearance of inhaled particles	Not a primary cell of origin; may contribute indirectly to squamous carcinogenesis via dedifferentiation or replacement by basal cells	
Bronchioalveolar stem cells (proposed)	Bronchioalveolar duct junction	Putative bipotent progenitor for bronchiolar and alveolar epithelium	Subset of adenocarcinomas (supported mainly by experimental models)	



Schematic shows goblet, club, ciliated, basal and neuroendocrine cells (PNEC), among others. BM: basement membrane; PFL: periciliary fluid layer; MGL: mucus gel layer ([contributed by Anthony Kicic](#)).



Autopsy lung specimen shows alveolar lumens covered by both flat cells (i.e., type I alveolar epithelium) and cuboidal cells (i.e., type II alveolar epithelium) ([contributed by Hirotsugu Hashimoto, M.D., Ph.D.](#)).

II. Epidemiology of lung cancer

Lung cancer is the [leading cause](#) of cancer related deaths in both men and women in the U.S. It is projected to cause [124,730 deaths](#) in 2025, including 64,190 in men and 60,540 in women, accounting for 20.2% of the total 618,120 cancer related deaths projected in 2025.

Lung cancer death rates have declined by 61% since 1990 in men and by 38% since 2002 in women, largely due to reductions in smoking. In addition, major advances in treatment for non-small cell lung cancer (better surgical techniques, targeted therapies, checkpoint inhibitors, stereotactic body radiotherapy, improved chemotherapy and improvements in multimodality care) and earlier detection (low dose CT screening for heavy smokers) have accelerated declines in the past decade. From 2013 to 2022, the lung cancer death rate decreased by 4.8% per year in men and 3.7% per year in women.

The 5 year relative survival rate for lung cancer is 27% overall, but it is 64% for patients with localized disease (27% of all cases), 36% for regional disease and 9% for metastatic (systemic) disease. For small cell lung cancer alone, the 5 year relative survival rate is only 5-10%.

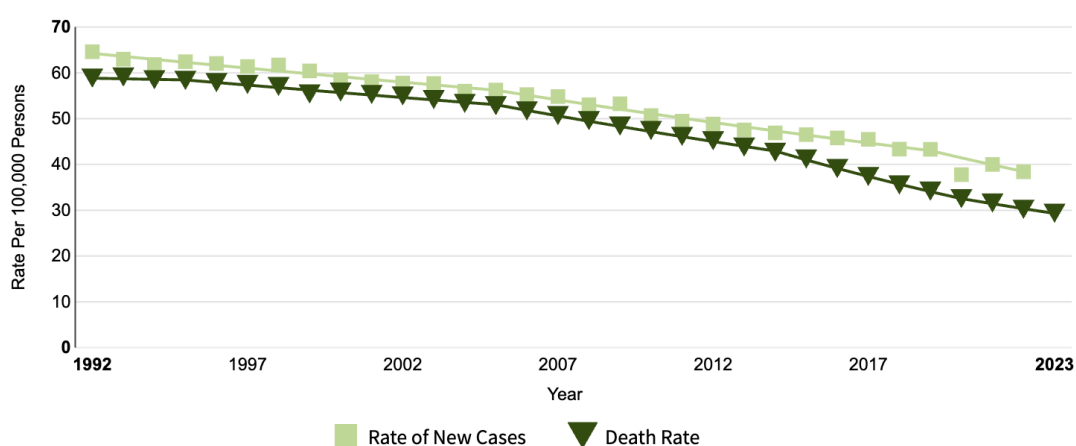
Lung cancer death rates, United States

Our World in Data

Reported annual death rate from lung, bronchus or trachea cancers per 100,000 people, based on the underlying cause listed on death certificates.



[Source: Our World In Data](#)



[Lung cancer cases and deaths - source: SEER \(2025\)](#)

III. Population attributable fraction for lung cancer

The most important lung cancer risk factor is tobacco smoking. Table 1 lists all major risk factors for lung cancer and their [population attributable fraction](#) (PAF) for smokers and nonsmokers. The population attributable fraction or population attributable risk refers to the proportion of cancer cases in a population that can be attributed to specific risk factors, both modifiable (tobacco smoking, secondhand smoke, radon exposure, occupational exposure) and nonmodifiable (random biological events, tuberculosis (TB), germline / family history, chronic obstructive lung disease, pneumonia).

The population attributable fraction for the same risk factor can vary substantially depending on the population studied. For example, the PAF for indoor air pollution varies among residents of the U.S. (low) and China (high due to the use of coal stoves for cooking). Some risk factors overlap, so PAFs are not additive.

We discuss lung cancer separately in smokers and never smokers based on striking differences in their epidemiological and molecular characteristics, clinical presentation and prognosis, which has led some authors to conclude they are [distinct clinical entities](#). Never smokers with lung cancer are more often women, more frequently of Asian/Pacific Islander or Hispanic ethnicity, have a higher frequency of adenocarcinoma, and more commonly harbor *EGFR* mutations and *ALK* rearrangements.

Never smokers account for [16% \(women\) and 11% \(men\)](#) of U.S. lung cancer deaths.

Table 1 - Population attributable fraction (PAF) of U.S. lung cancer cases in smokers and never smokers

Risk factor	Smokers	Nonsmokers
Tobacco (smoking)	75%	not applicable
Random biological events	not available	50%
Radon	10%	20%
Occupational	5-15%	not available
Secondhand smoke	not available	10%
Outdoor air pollution	2%	not available
Germline / family history	not available	2%
Tuberculosis	not available	1-2%

No traditional PAF calculated: aging

PAF < 1% in U.S.: chronic obstructive lung disease (COPD), household use of coal / indoor air pollution

No clear association: cannabis, diet, HIV, HPV, pneumonia

Lung cancer risk factors

Tobacco (smoking)

Between 1991 and 2018, the fraction of U.S. lung cancer deaths attributable to cigarette smoking declined from [81 percent to 75 percent](#). During this period, lung cancer mortality attributable to smoking fell by about 2.7 percent per year. Lung cancer risk increases with both the quantity and duration of smoking and risk decreases after quitting, with progressive risk reduction the longer a person remains abstinent.

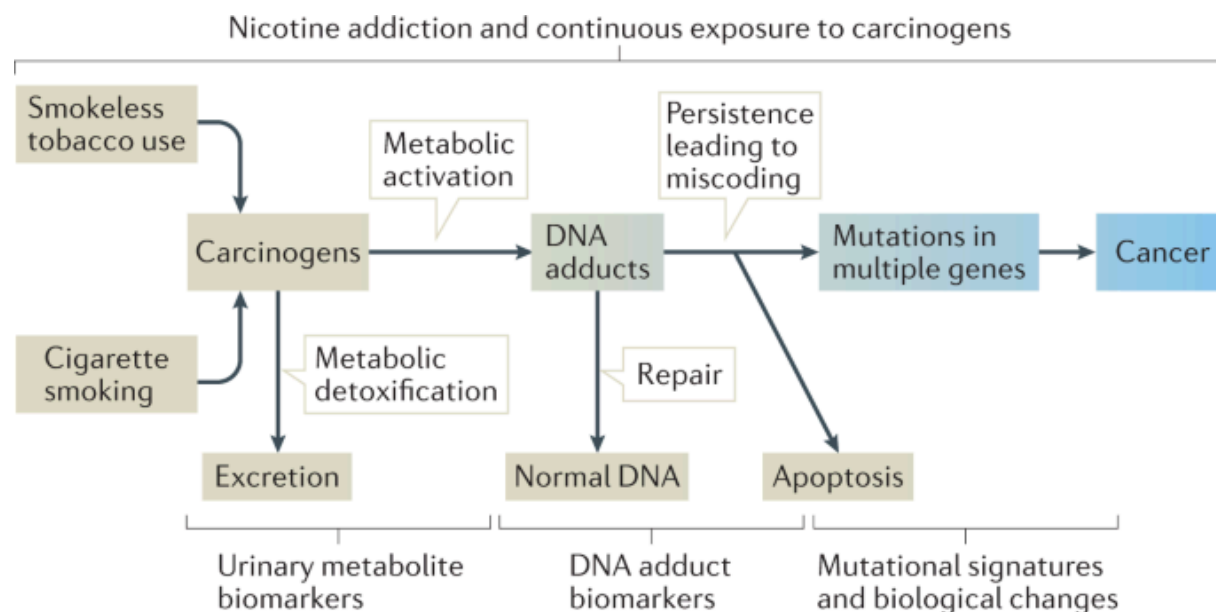
Cigar and pipe smoking can produce lung cancer risks comparable to cigarette smoking when inhalation is deep and use is long term, although reliable population attributable fractions are difficult to estimate due to low prevalence, heterogeneity of use and frequent use alongside cigarettes. E-cigarettes, which may deliver nicotine in aerosol form without burning tobacco, have not yet been definitively linked to human

lung cancer; however, they can generate carcinogenic compounds during heating and are [discouraged by public health authorities](#) because of toxic exposures and limited long term safety data.

Tobacco smoking causes lung cancer through multiple mechanisms:

Tobacco carcinogens cause DNA mutations

Tobacco smoke contains more than 70 established carcinogens, including polycyclic aromatic hydrocarbons such as benzo[a]pyrene, tobacco specific nitrosamines (NNK and NNN), aromatic amines, aldehydes and volatile substances such as 1,3-butadiene and benzene. Many undergo metabolic activation to reactive intermediates that form DNA adducts. This contributes to a characteristic pattern of somatic mutations, including an excess of C→A transversions in lung adenocarcinoma and squamous cell carcinoma, especially in *TP53* and *KRAS* genes.



Contributed by [Hecht and Hatsukami](#)

Carcinogens may also undergo metabolic detoxification, often through conjugation with glutathione or glucuronic acid mediated by glutathione S transferases and UDP-glucuronosyltransferases and sulfotransferases. These processes can render metabolites more water soluble and easier to excrete, but intermediate products may remain reactive.

Smoking increases the total mutational burden and the number of driver mutations in lung cancers, with mutation levels that can exceed those seen in nonsmokers by more than 10 fold. High mutation loads may impair DNA repair pathways and accelerate tumor evolution.

Smoking also causes [field cancerization](#), characterized by expanded regions of genetically heterogeneous premalignant epithelial clones that may give rise to multiple independent tumors.

Smoking causes chronic inflammation

Cigarette smoke induces chronic inflammation in the airways, with infiltration by neutrophils and macrophages and the generation of reactive oxygen species. These promote oxidative DNA damage, lipid peroxidation and selection of clones with survival advantages. The resulting microenvironment facilitates tumor progression and accumulation of additional mutations.

Additional mechanisms

Smoking may also:

- impair DNA repair because some aldehydes in smoke block DNA repair enzymes and suppress nucleotide excision repair.
- promote proliferation of mutated cells through nicotine and nitrosamines acting on growth signaling pathways.
- induce epigenetic alterations, including promoter hypermethylation (such as CDKN2A silencing), global hypomethylation and dysregulation of microRNAs.
- suppress immune surveillance by altering antigen presentation, impairing dendritic cell maturation and promoting T cell exhaustion.

Genetic susceptibility

Germline [variants in nucleotide excision repair genes](#) and other pathways (such as *CYP1A1*) involved in handling bulky DNA adducts may increase susceptibility to lung cancer in smokers. Associations may vary by population and may contribute disproportionately to the risk of squamous cell carcinoma. Genetic variation affecting activation and detoxification pathways also appears to influence susceptibility to smoking related cancer.

Complex Interactions

The biological effects of tobacco smoke reflect the combined actions of an estimated [7,000 chemicals](#), interacting with diverse genetic, epigenetic, inflammatory and immune pathways. Because of this panoply of interactions, predicting the impact of long term exposure at the individual level remains difficult. Rather than acting through a single dominant mechanism, tobacco smoke drives carcinogenesis through cumulative disruption of multiple networks controlling genomic integrity, cell growth, immune function and tissue homeostasis.

Random biological events

Lung cancer in never smokers, defined as those who have smoked fewer than 100 cigarettes in their lifetime, accounts for [10-20% of annual lung cancer cases](#) in the United States. Mortality rates may be slightly lower because never smokers often have molecular subtypes with better treatment responses. Many researchers consider lung cancer in never smokers to be a biologically [distinct disease](#) from that arising in smokers. If treated as a separate category, it would rank among the [top ten causes](#) of cancer death in the United States.

In this paper, we attribute a substantial proportion of lung cancers in never smokers to causes that are not currently identifiable, which we term “random biological events” that have no definite association with known risk factors in a particular patient. This includes errors during [DNA replication](#) in noncancerous stem cells, errors in chromatin organization or [epigenetic](#) regulation, errors in the [segregation of cellular components](#) during mitosis and patient specific immune system failures that are ineffective at eliminating premalignant or malignant clones.

We estimate that this category may account for half of lung cancer cases in never smokers in North America and Europe. This estimate is calculated by subtracting the attributable risk of established environmental and occupational exposures from 100% (the total burden of disease); it represents an approximation rather than a formal population attributable fraction due to factors such as limitations in exposure measurement, confounding variables and risk factor classification.

We believe there may be a baseline rate of lung cancer, which we estimate as 2 cases per 100,000 per year, compared with the current [age adjusted U.S. incidence](#) of lung cancer of 47.8 cases per 100,000 per year. This estimate is based on the [lowest age adjusted lung cancer rates](#) reported globally, such as in parts of Sub-Saharan Africa, although these rates may be inaccurate due to underdiagnosis, limited surveillance and lower life expectancy. This baseline is also within the range of historical U.S. lung cancer rates before widespread tobacco use, which have been reported as [less than 5 per 100,000](#), although comparisons must be made cautiously due to changes in diagnostic practices.

Radon exposure

Indoor radon is a naturally occurring radioactive noble gas generated by the decay of radium within the uranium decay series. It is colorless, odorless and tasteless and is classified as a [Group 1 carcinogen](#) by the International Agency for Research on Cancer. Indoor radon concentrations vary widely by geographic region and are influenced by underlying geology, building characteristics and ventilation practices.

Epidemiologic studies, including pooled residential case control analyses, demonstrate that indoor radon exposure increases lung cancer risk in both smokers and never smokers. Radon is the leading environmental cause of lung cancer in individuals who have never smoked. Estimates suggest that [20%](#) of lung cancer deaths among never smokers and [10% among ever smokers](#) are attributable to residential radon exposure, with variation depending on population smoking prevalence, exposure distributions and modeling assumptions (see also [here](#)). Although the attributable fraction is higher in never smokers, the majority of radon attributable lung cancer deaths occur among current or former smokers because of the substantially higher baseline risk in these groups.

Radon itself is chemically inert and a large proportion is exhaled after inhalation. Lung cancer risk arises primarily from [short lived radon decay products](#), including polonium 218, lead 214, bismuth 214 and polonium 214, which attach to inhaled aerosols and deposit along the bronchial epithelium. These radionuclides emit alpha particles that deliver high linear energy transfer radiation to epithelial cells.

Alpha particle irradiation produces dense ionization tracks that result in complex DNA damage, including double strand DNA breaks, clustered lesions and chromosomal aberrations. Indirect effects mediated by reactive oxygen species may also contribute. The resulting genomic damage is often difficult to repair and can promote malignant transformation.

Radon exposure acts [synergistically with tobacco smoking](#) to increase lung cancer risk. Experimental, epidemiologic and modeling studies support several interacting mechanisms, including combined DNA damage from alpha radiation and tobacco derived carcinogens, altered cellular responses to DNA damage and chronic inflammation with increased epithelial turnover that facilitates fixation of genetic alterations.

Occupational exposure, including asbestos

Lung cancer is the leading malignancy attributable to occupational exposures, with estimates suggesting that [5-15% of cases](#) in the U.S. population may result from occupational agents. Variability in estimates reflects differences in exposure prevalence, definitions, study methods and latency between exposure and diagnosis. [Well established occupational carcinogens](#) include asbestos, arsenic, beryllium, cadmium, chromium, coal-tar products, diesel exhaust, nickel, polycyclic aromatic hydrocarbons, silica, soot and welding fumes. Radon can also be an occupational risk, particularly in mining or underground work.

Asbestos refers to naturally occurring fibrous silicate minerals, classified into serpentine (chrysotile) and amphibole types, with amphiboles being generally more biopersistent and carcinogenic. Cigarette smoking [amplifies the carcinogenic effects](#) of asbestos. In studies of [insulation workers](#), lung cancer mortality among nonsmokers exposed to asbestos was elevated and risks were highest among smokers with asbestos exposure, suggesting more than additive effects.

Occupational carcinogens likely act through multiple mechanisms. Asbestos fibers induce chronic inflammation, generate reactive oxygen species and can cause direct genotoxic effects. They may also [modulate immune function](#), contributing indirectly to tumorigenesis. Interactions between tobacco smoke and asbestos, including overlapping inflammatory pathways and iron catalyzed [oxidative stress](#), may underlie their [combined effects](#).

Data on occupational risk in never smokers [are limited](#) and precise population attributable fractions cannot currently be assigned for this group.

Secondhand smoke

Secondhand smoke, also referred to as environmental tobacco smoke (and historically as passive smoking), is an established cause of lung cancer. The [U.S. Surgeon General](#) estimated that during 2005-2009, 4.6% of lung cancer deaths in U.S. adults aged 35 years or older were attributable to secondhand smoke, with the burden occurring predominantly among never smokers. In [Alberta, Canada](#), secondhand smoke exposure was estimated to account for 5.2% of lung cancer cases among never smokers in 2012.

Among never smokers, population attributable fractions (PAFs) for lung cancer due to secondhand smoke vary substantially by region, reflecting differences in the prevalence and intensity of long-term household and workplace exposure. Using relative risk estimates summarized by [Sisti](#), model derived PAFs in never smokers were 8-12% in men and 6-24% in women across North America, Europe and China. [European cohort studies](#) have reported higher overall PAFs, reaching 16–24%, particularly in populations with substantial domestic or occupational exposure. For this review, a value of 10% was used for never smokers in Table 1 as a conservative pooled estimate.

Among active smokers, the additional contribution of secondhand smoke exposure is difficult to interpret at the population level. Although incremental risk can be modeled, the dominant effect of direct smoking limits the meaningful estimation of a separate attributable fraction for passive exposure.

Secondhand tobacco smoke consists primarily of sidestream smoke emitted from the burning end of a cigarette, combined with diluted mainstream smoke exhaled by the smoker. The [International Agency for Research on Cancer \(IARC\)](#) classifies secondhand smoke as a Group 1 carcinogen. In never smokers, chronic exposure, particularly cohabitation with a smoker, is associated with a [20-30% increase](#) in lung cancer risk. Carcinogenic effects arise through mechanisms similar to those observed in active smokers, including DNA adduct formation, oxidative injury and carcinogen induced mutagenesis.

Outdoor air pollution

In 2013, the International Agency for Research on Cancer classified outdoor air pollution as a whole and particulate matter, particularly [PM2.5](#), as [Group 1 carcinogens](#) with sufficient evidence for causation of lung cancer. Estimates by [Doll and Peto](#) in the 1980s suggested that 1-2% of lung cancers were attributable to air pollution, reflecting both lower ambient exposures at the time and substantial methodological limitations in exposure assessment. [Subsequent studies](#) using improved monitoring and modeling have generally yielded higher estimates.

In [Alberta, Canada](#), between 1.9% and 5.7% of incident lung cancer cases in 2012 were attributed to long term exposure to outdoor PM2.5. Global modeling studies estimate that [10-15% of lung cancer](#) cases or deaths worldwide are attributable to PM2.5 exposure, with substantial regional variation driven by differences in ambient pollution levels and population susceptibility. For Table 1, a conservative estimate of 2% was used to represent the contribution of outdoor air pollution to lung cancer in the United States. This estimate reflects generally lower average PM2.5 concentrations in the U.S. compared with global levels and assumes a broadly similar exposure profile to Canada, while acknowledging marked regional heterogeneity.

Outdoor air pollution contributes to lung carcinogenesis through chronic inhalational exposure to complex mixtures of particulate matter and gaseous pollutants, including ozone, nitrogen dioxide and combustion derived particles. PM2.5 promotes [oxidative stress](#) and persistent pulmonary inflammation and serves as a carrier for carcinogenic compounds, leading to DNA damage, DNA adduct formation and mutagenesis. Polycyclic aromatic hydrocarbons, including [benzo\[a\]pyrene](#), are among the best characterized carcinogenic constituents of air pollution and can induce mutational signatures and epigenetic alterations that may contribute to lung cancer initiation and progression.

Germline variations / family history

Familial aggregation of lung cancer reflects both inherited susceptibility and shared environmental exposures, particularly smoking. Pooled analyses indicate that individuals with a first degree relative affected by lung cancer have a [1.5 to 2.5 times](#) higher risk after adjustment for smoking and other

confounders, with the strongest association observed when the affected relative is a sibling. High risk germline variants [have been detected](#) in a small subset of patients with a documented family history of lung cancer (see also [here](#) and [here](#)). In lung adenocarcinoma, [2.5 to 4.5% of patients](#) carry pathogenic or likely pathogenic variants in DNA repair pathway genes such as *ATM*, *TP53*, *BRCA2* and occasionally *EGFR* - these variants are individually rare in the general population but may contribute to familial clustering.

A [very rare syndrome](#) involving germline *EGFR* T790M mutations has been described, predominantly in never smokers. Family based studies suggest that carriers of this variant may have an approximate 31% lifetime risk of developing lung cancer, although this estimate is uncertain due to very small sample sizes and potential ascertainment bias. Risk appears largely restricted to adenocarcinoma.

Among never smokers, [Sisti estimated](#) that 1.2–2.9% of lung cancers may be attributable to family history, with geographic variation reflecting both differences in genetic background and shared environmental exposures (2.0 percent in North America, 1.2 percent in Europe and 2.9 percent in China). For Table 1, a conservative population attributable fraction of 2% was used for never smokers. Much of the familial risk remains unexplained, reflecting additional genetic and environmental factors yet to be identified.

Tuberculosis

Tuberculosis (TB) has been associated with an increased risk of lung cancer. Epidemiologic studies suggest that the population attributable fraction (PAF) of lung cancer due to tuberculosis varies by region and baseline TB prevalence. [Leung estimated](#) that in high income areas of North America, 4.8% of lung cancers in smokers and nonsmokers combined were attributable to tuberculosis, compared with 18.1% in Southeast and Eastern Asia. Among never smokers, model based estimates by [Sisti](#) suggest attributable fractions of 1.1% in North America, 2.4% in Europe and 12.7% in China. For this review, a conservative PAF of 1-2% was used for never smokers in Table 1, reflecting both the limited available data and regional differences in TB prevalence.

Tuberculosis may promote lung carcinogenesis through several mechanisms. Chronic inflammation generates reactive oxygen and nitrogen species, leading to DNA damage. Repeated cycles of injury and repair increase the mutational burden in bronchial and alveolar epithelial cells. Cytokine mediated signaling may enhance the survival and proliferation of damaged cells, while [pulmonary scarring](#) creates a microenvironment conducive to malignant transformation. These processes likely act over many years, contributing predominantly to peripheral lung cancers in areas of prior TB involvement.

Aging

Aging is the strongest risk factor for lung cancer, with a [median age](#) of onset of 70 years. Possible mechanisms include:

- Somatic mutations [accumulate with age](#), increasing cancer susceptibility.
- Age related changes in cellular networks and tissue function may interact with other risk factors.
- Aging is associated with inflammaging, a chronic, low grade systemic inflammation with elevated cytokines (e.g., IL6, TNF- α , CRP) and [immune system dysfunction](#).
- [Epigenetic alterations](#) occur with age, potentially affecting chromatin stability and gene expression.

A [recent study](#) estimated the population attributable fraction of lung cancer deaths due to aging at 78%. However, this is not included in Table 1 because these calculations typically do not involve biological processes.

Risk factors with a PAF < 1% or that cannot be determined

Cannabis smoking: The link between cannabis (marijuana) smoking and lung cancer [remains inconclusive](#), with conflicting epidemiological evidence.

Chronic obstructive pulmonary disease (COPD): COPD is a strong independent risk factor for lung cancer but calculations are confounded by smoking, which is present in most COPD patients. However, in never smokers in North America, [Sisti](#) attributed 0.4% of lung cancer cases to COPD compared with 0.6% in China; no estimate was provided for European populations. COPD is characterized by an [excessive inflammatory](#) response and [dysregulated immune function](#) that may contribute to its association with lung cancer.

Diet: In the U.S., diet is so [closely confounded with smoking](#) that it is difficult to discern an independent effect. For these reasons, [Sisti](#) chose not to assess the impact of diet on the risk of lung cancer.

HIV: There are no reports on a PAF of HIV and lung cancer, although HIV patients have a 1.5-3 fold [increased risk of lung cancer](#) and lung cancer is the most frequent nonAIDS defining cancer. The increased risk may be due to immunosuppression, recurrent lung inflammation, impaired repair or interaction with other risk factors, such as smoking.

Household combustion of coal: The [IARC has concluded](#) that this use, more common in [low and medium resource countries](#), causes lung cancer. [Sisti](#) attributed 19.9% of lung cancer cases in Chinese women who were never smokers to household use of coal. Burning coal, particularly with poor ventilation, generates respirable particles and many carcinogens including benzo[a]pyrene, formaldehyde and benzene. However, this is not a significant factor in the U.S.

HPV: There are contradictory reports that [HPV does](#) or [does not](#) increase the risk for lung cancer. There also does not appear to be any conclusive study providing a PAF for HPV and lung cancer.

Pneumonia: A history of pneumonia [may modestly increase lung cancer risk](#), but there is no consensus on the PAF for the association. The association may be due to inflammation that continues even as the clinical symptoms resolve.

Risk factors interact with each other

Risk factors often interact because a single behavior can influence multiple risks (smoking and diet), risk factors can have synergistic effects (smoking and asbestos exposure) and the biological mechanisms underlying different risk factors may overlap.

IV. How life arose and our biological structure

How life arose

Life is thought to have arisen more than [3.5 billion years ago](#) through chemical processes that produced self sustaining, self replicating systems capable of metabolism and heredity. As emphasized by [Kauffman](#) and others, networks of [highly reactive](#), interacting biomolecules likely played an important role, although the precise sequence of events that led to the first living cells remains unresolved. Early single celled organisms likely emerged from [self organizing](#) networks that integrated metabolism, survival and replication rather than being “programmed” solely for reproduction.

Multicellularity emerged multiple times during evolution. Simple multicellular organisms appeared more than [1.7 billion years ago](#), while complex multicellular life arose much later, roughly 600 million years ago. Multicellular organisms built on the molecular and regulatory framework of single cells but evolved additional mechanisms to coordinate cell differentiation, tissue organization and the timing of cell division across many cells.

Living systems rely on interconnected networks of genes, proteins and metabolites. Core networks, such as those governing cell division and energy metabolism, are [reused](#) in different contexts but produce different outcomes depending on cell type, developmental stage and environmental conditions. Tight regulation of when and where these networks are active is essential for normal physiology. Because many biomolecules can interact with multiple partners, regulatory mechanisms are required to ensure specificity and coordination. Over evolutionary time, natural selection has refined these regulatory systems to enhance survival and reproductive success.

Unlike most machines, which can be explained by [reductionist principles](#) (i.e., the behavior of the whole equals the behavior of the sum of the parts), human life is a [complex adaptive system](#) made up of thousands of interacting biological networks, shaped by both their structure and their history. In such systems, the [behavior of the whole](#) is greater than the sum of the behavior of the individual parts because of the interactions between the parts. In humans, [20,000 protein coding genes](#) give rise to gene products that interact through regulatory networks involving transcription factors, signaling molecules, chromatin modifiers and noncoding RNAs. These networks control the production, modification and localization of biomolecules that guide development from a fertilized egg through all stages of life. They also enable responses to environmental challenges such as infection, injury and temperature extremes and they provide the biological substrate on which evolution acts at the population level.

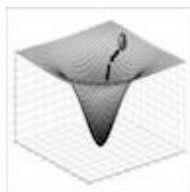
Human development begins with a fertilized egg that uses maternally supplied [proteins and RNA transcripts](#) to initiate rapid cell divisions and early patterning. As development proceeds, cells produce and respond to signaling molecules, including [morphogens](#), that form concentration gradients across developing tissues. These gradients activate different sets of genes in different cells, leading to stable [differences in gene expression](#) and cell fate despite a shared genome. Although differentiation is often progressive, it is not strictly irreversible and depends on ongoing regulatory input.

Gene regulation frequently occurs through hierarchical networks in which the activation of one gene influences the expression of many others through its gene products. [Homeobox genes](#) are a well known example. They encode transcription factors that specify positional identity along body axes and regulate downstream gene expression programs. These programs in turn control cellular behaviors such as proliferation, migration and differentiation, ultimately shaping body plans and anatomical structures.

The origin of order

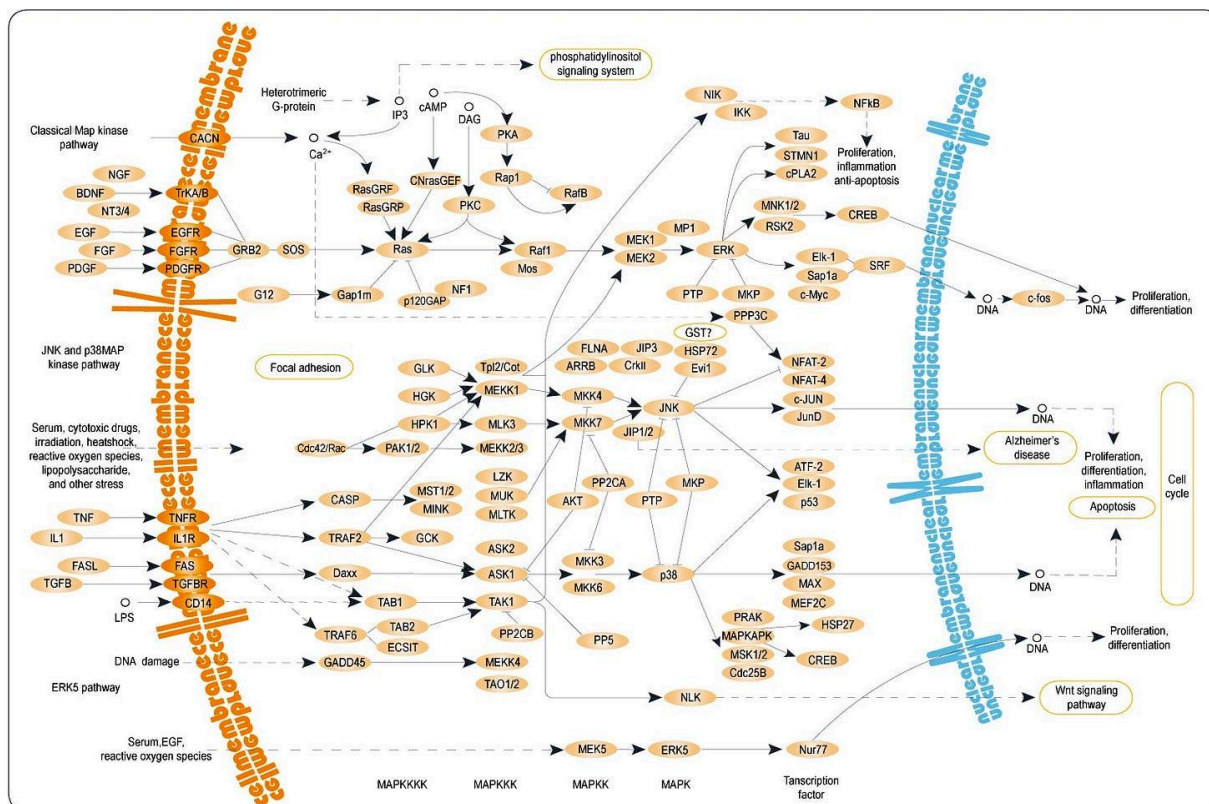
[Kauffman proposed](#) that complex systems, including living organisms, can spontaneously generate structured patterns of organization through interactions among their components. This emergent order provides a framework for understanding the development and stability of life on Earth.

Biological order is reinforced by evolutionarily selected mechanisms. In humans, redundant control systems have evolved that constrain the inappropriate activity of biomolecules and networks. For example, multiple checkpoints and regulatory pathways limit uncontrolled cell proliferation. However, evolution operates over long timescales (up to [1 million years to alter a trait](#)) and rapid changes in environmental or lifestyle factors can outpace the development of protective traits. In addition, natural selection acts most strongly on [traits that affect reproductive success](#), which may partly explain why mechanisms protecting against late onset diseases, such as cancer, are relatively weak.



Cell attractors are often modeled as a valley in a topographic diagram.

Order is also maintained by [cell attractors](#), which pull cells into a [common configuration](#). Attractors can be thought of as [low energy states](#) or valleys in a topographic landscape that stabilize cellular networks against common disruptions. Just as a ball dropped into a valley tends to remain there, cellular networks within an attractor state resist change. These states emerge from the countless interactions among networks that buffer against perturbation and [resist change](#). This concept helps explain how the body's 30 trillion cells, despite subtle variations in network activity, [converge](#) into only about [230 distinct cell types](#).



[A complex web of interactions](#) for the MAP kinase pathway, related to cell division.

Networks not molecules

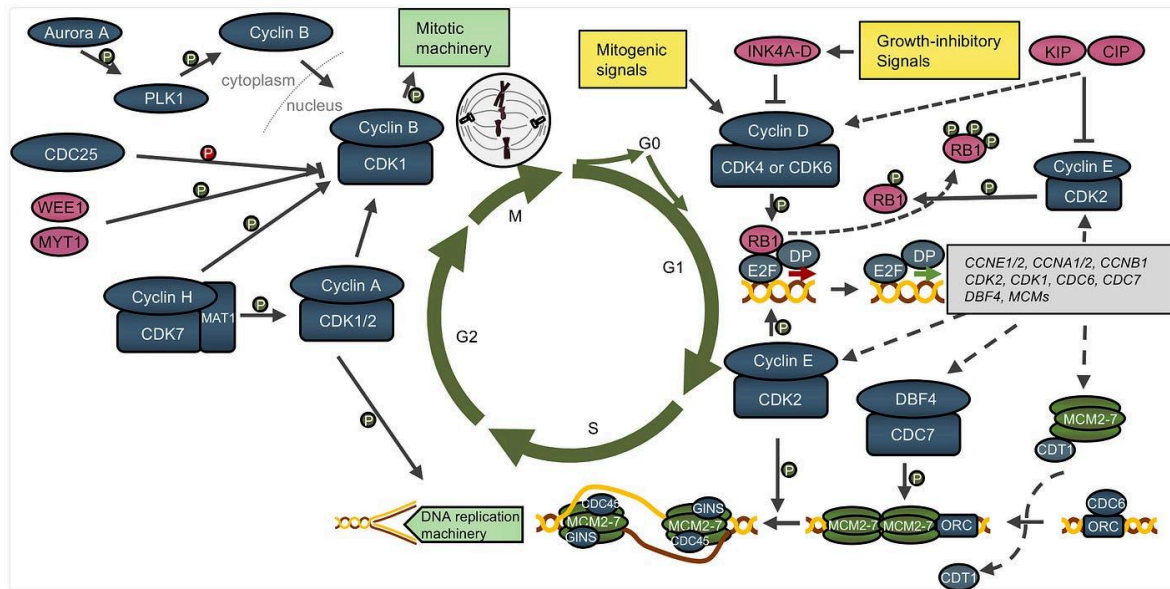
Life is better understood by analyzing the behavior of large scale biological networks in addition to individual molecules or linear pathways. While pathways are often presented in textbooks as stepwise sequences with defined start and end points, in reality, they are interconnected with many other pathways, forming complex networks and a dense, interconnected web. Disruptions in one pathway can sometimes be buffered by alternative routes, although critical nodes may remain essential for proper function. This perspective [suggests](#) that changes in overall network behavior may provide deeper insights than focusing only on specific genetic mutations.

Organic codes

Cell function depends not only on DNA but also on organic codes and cellular memory. [Organic codes](#) are environmental signals, such as physical forces or mechanical stress, that cells interpret according to their own internal rules. For example, proteins fold into specific shapes based on these codes to optimize function and limit harmful interactions, a process not strictly encoded in DNA.

Cellular memory refers to heritable regulatory states, including [histone](#) modifications, DNA methylation and other [epigenetic](#) marks, that persist through cell division and influence gene expression without altering the DNA sequence. Together, these mechanisms allow cells to maintain identity, respond to environmental changes and regulate their behavior beyond the information directly encoded in the genome.

Figure 1. Overview of the cell-cycle.



A [simplified version](#) of the cell cycle shows numerous levels of network involvement. Each of these genes and proteins is also controlled by a web of other genes and proteins, which makes the overall flow difficult to understand.

The sophistication of cell division

Human cell division is a highly complex and tightly regulated process involving multiple coordinated molecular and structural events. Rather than it being a linear sequence, it is controlled by a network of interacting regulators that ensure accuracy and timing, resembling a multidimensional web.

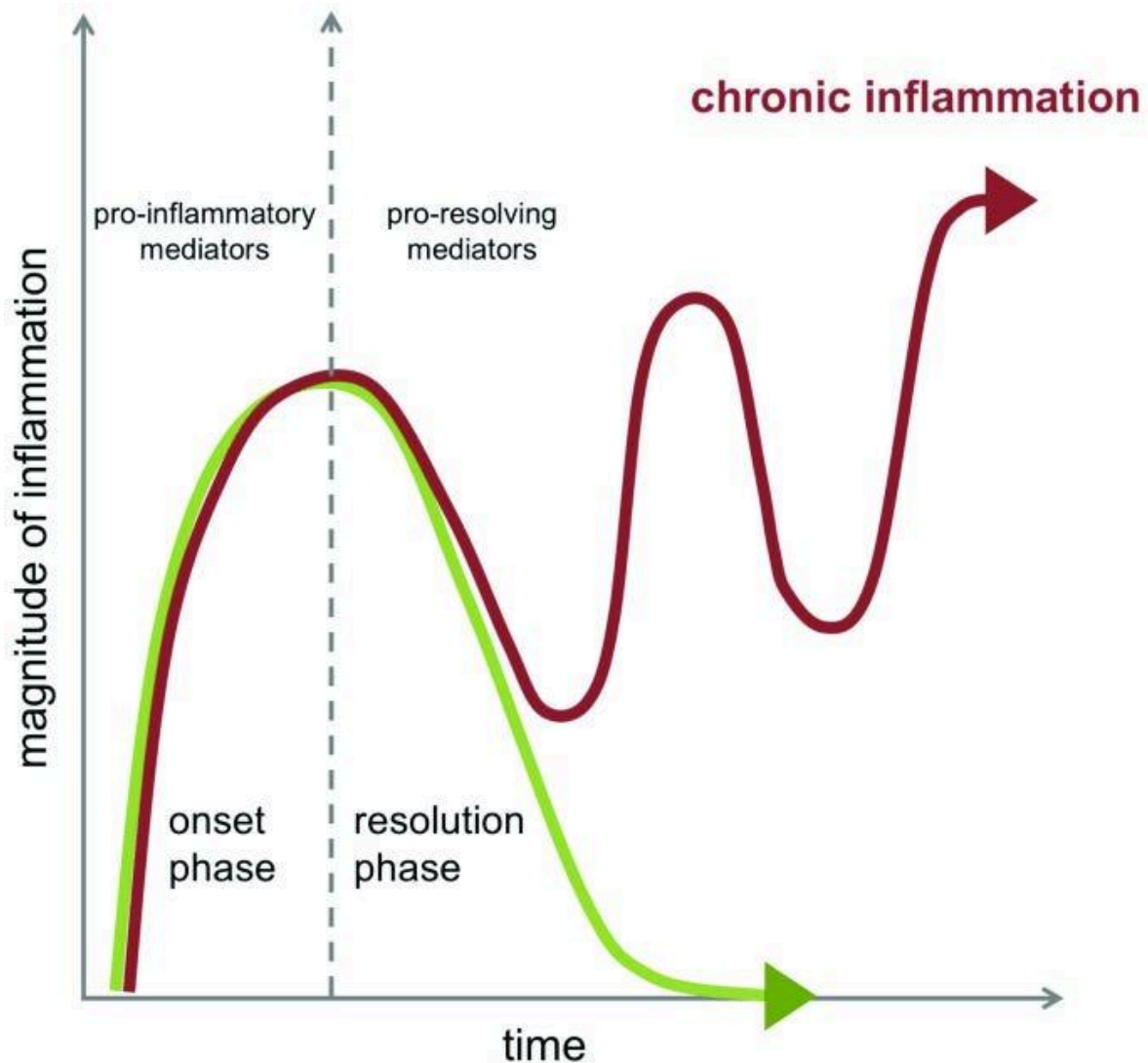
Cells progress through the cell cycle, which includes precise duplication of the genome followed by mitosis (nuclear division) and cytokinesis (cytoplasmic division). [Replication](#) of DNA's 3 billion base pairs is central to cell division and high fidelity is essential. Over [200 genes](#) contribute to DNA repair, reducing the error rate from 1 per 100,000 to [1 per 100 million](#) base pairs. Even so, an average of 30 errors may occur per genome per replication, though many are corrected by post replication repair mechanisms.

- Cyclin dependent kinases (CDKs) and their regulatory cyclins are cyclically activated to coordinate progression through the cell cycle. These kinases integrate internal and external signals and their activity is modulated by multiple regulatory mechanisms.
- Cell cycle checkpoints at G1/S and G2/M monitor DNA integrity and spindle assembly. When errors are detected, these checkpoints can halt progression to allow repair or activate signaling pathways, such as those involving p53, that trigger programmed cell death if damage is irreparable. Feedback and feedforward loops reinforce the proper timing of cell cycle transitions, while CDK inhibitors directly bind cyclin-CDK complexes to prevent premature or inappropriate progression.
- Extracellular signals, including growth factors, regulate cell cycle entry and exit by modulating cyclin and CDK expression and activity.

Mitosis requires extensive remodeling of the cytoskeleton and membranes, assembly of the mitotic spindle, disassembly of the nuclear envelope and redistribution of organelles to ensure equal division between daughter cells.

Coordination of network activity

Complex biological processes, such as inflammation and embryogenesis, require coordinated activity among multiple interacting networks. These networks operate according to regulated patterns rather than a predetermined plan.



[The processes of inflammation](#) (red line) and resolution of inflammation (green line) begin simultaneously; if the resolution is not effective, inflammation continues indefinitely.

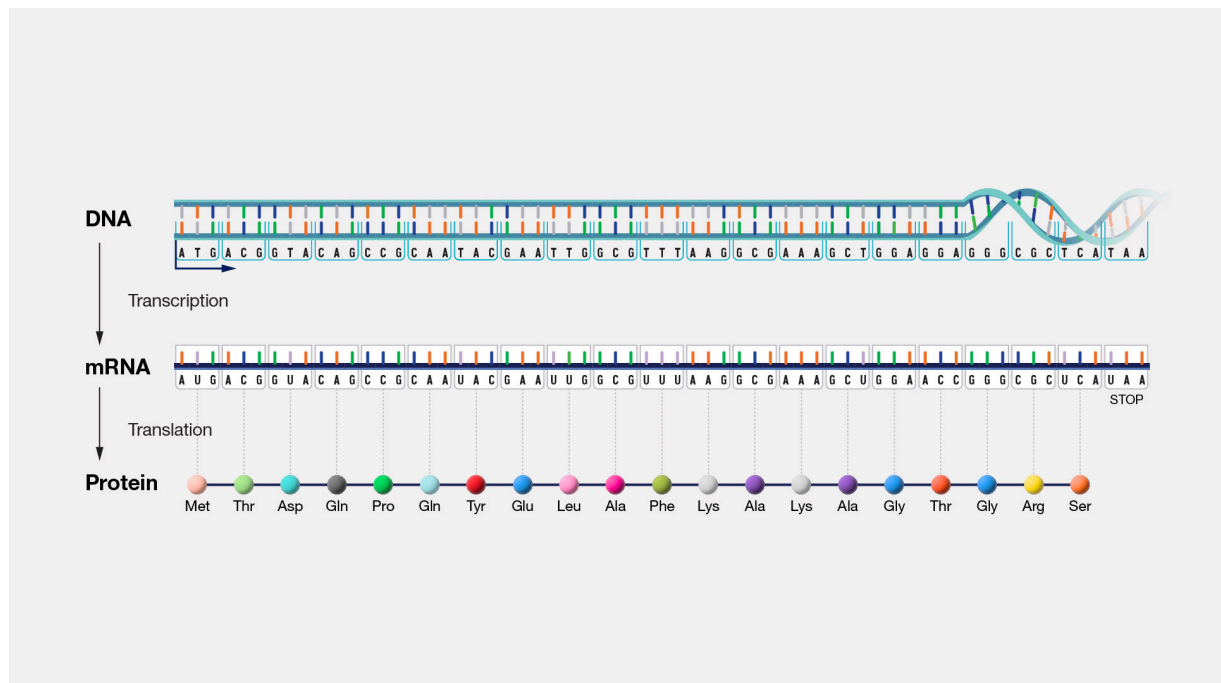
In the inflammatory response, pathways that initiate inflammation are closely [coordinated](#) to those that promote [resolution](#). Early inflammatory signals recruit immune cells, increase vascular permeability and eliminate pathogens or damaged tissue. Resolution pathways are typically activated shortly thereafter and restore cellular networks to their baseline state once the injury or microbial threat is removed, limiting [collateral tissue damage](#). In contrast, inflammation triggered by cancer risk factors [does not](#) properly activate these resolution mechanisms, leading to chronic inflammation that promotes genomic instability and fuels malignant progression.

Embryogenesis exhibits processes that resemble features of malignancy but are tightly controlled and [halted during the fetal stage](#). These include:

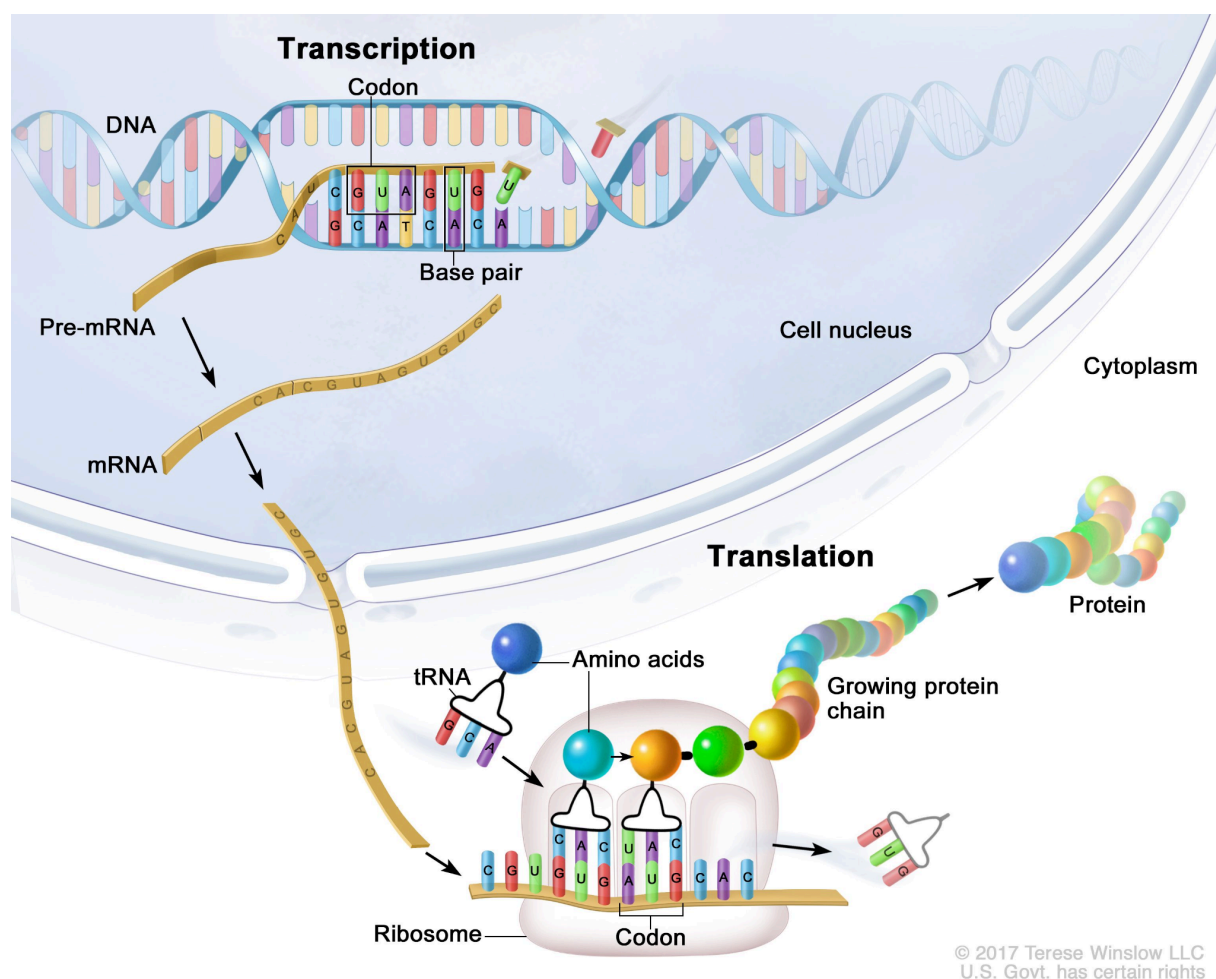
- [Rapid cell division](#), which ends at the transition from maternal mRNA driven control to [embryonic](#) genome-driven control of gene expression.
- [Morphogenesis](#), guided by [asymmetric](#) morphogen gradients and [self organizing patterns](#) of signaling molecules that direct tissue and organ formation.
- [Cell differentiation](#), in which progenitor cells adopt specialized identities.
- [Cell migration](#), allowing cells of different lineages to move short and long distances to establish the developing body plan.

Together, these examples illustrate how the coordination of network activity underlies both normal physiological processes and the controlled execution of complex developmental programs.

DNA transcription and translation



The [central dogma](#) of molecular biology describes the flow of genetic information from DNA to RNA through transcription and then from RNA to protein through translation. While some exceptions exist, such as RNA viruses that can produce proteins directly from RNA, the overall process is far more complex than typical textbook diagrams suggest. Successful transcription and translation depend on a highly coordinated network of biomolecules and regulatory factors operating within each active cell.



[National Cancer Institute](https://www.nationalcancerinstitute.gov)

Transcription

Transcription is a highly regulated process involving multiple stages: initiation, elongation and termination, all based on the DNA template. Each stage requires specific protein complexes, nucleic acids and regulatory molecules that control interactions at each step. Key components include:

- **Transcription factors:** Proteins that assemble into a [preinitiation complex](#) of 100 proteins to facilitate RNA polymerase recruitment.
- [Transcriptional coactivators:](#) Molecules that transmit regulatory signals from transcription factors to RNA polymerases.
- **Gene specific transcription factors:** Proteins that bind enhancers or silencers to modulate initiation of individual genes, often assisted by cofactors and coactivators.
- **Chromatin remodelers and histone modifiers:** Enzymes that alter chromatin structure to regulate access to the DNA template.
- **Regulatory RNAs:** Small RNAs that assist in fine tuning transcription and maintaining genomic stability.

This complex interplay ensures precise gene expression and allows cells to respond to environmental and developmental cues.

Translation

Translation, the synthesis of proteins from mRNA, is equally sophisticated and occurs in four main stages: [initiation, elongation, termination and ribosome recycling](#). Each stage relies on numerous biomolecules and cofactors:

- **Initiation:** Assembly of the ribosome on mRNA, recruitment of initiator tRNA and recognition of the start codon.
- **Elongation:** Sequential addition of amino acids to the growing polypeptide chain, delivered by tRNAs and charged by aminoacyl tRNA synthetases.
- **Termination:** Recognition of the stop codon and release of the completed polypeptide.
- **Ribosome recycling:** Disassembly of ribosomal subunits for reuse in future rounds of translation.

Additional essential components include ribosomal proteins, translation factors (initiation, elongation, release and recycling factors), energy sources (ATP and GTP), surveillance mechanisms that ensure fidelity and chaperone proteins that assist in proper folding of newly synthesized polypeptides.

Normal cellular physiology depends on the coordinated activity of these and other interacting processes. Sophisticated regulatory features and multiple redundant controls minimize the impact of errors so that disruptions in a small number of components rarely lead to significant failure. The precision and resilience of transcription and translation illustrate the complexity of gene expression and the importance of networked regulation in maintaining cellular function.

V. How cancer arises, general overview

Cancer is defined as the [uncontrolled proliferation of transformed cells subject to evolution by natural selection](#). It develops through chronic disruption of biological networks. The process is initiated by cancer risk factors and random genetic errors that weaken the redundant control systems of these networks, although clinical cancer usually takes years to decades to emerge.

Cancer seems to arise through two main pathways:

- In children and young adults, inherited germline variants can impair a few critical control mechanisms, making malignancy more likely once additional random biological events occur.
- In adults, long term exposure to cancer risk factors gradually damages multiple components of these networks, eventually overwhelming their control systems and permitting uncontrolled cell growth. The specific features of each malignancy depend on which cells are affected and which control mechanisms fail.

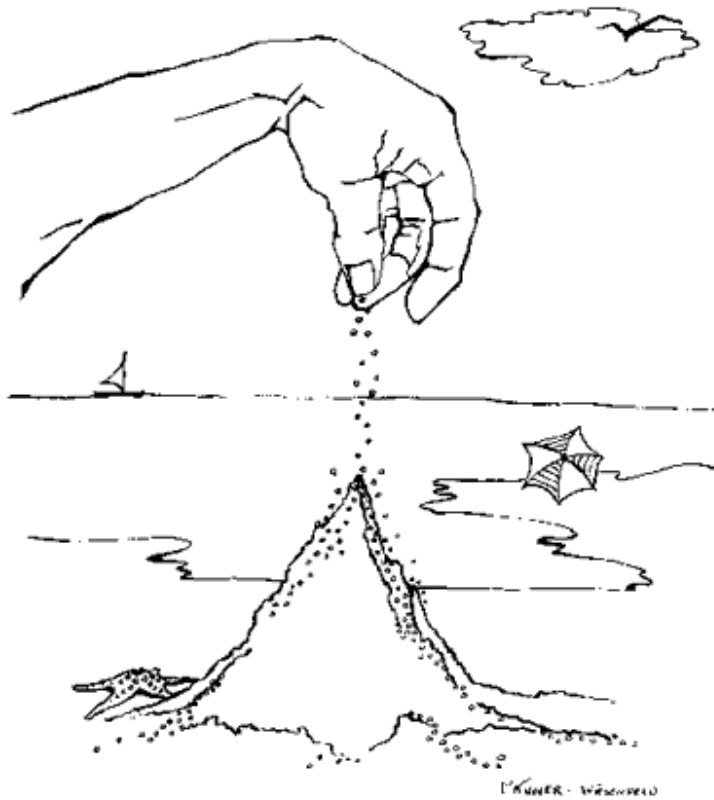
All human cells contain the same DNA, yet their appearance and behavior differ because gene expression is influenced by factors beyond the DNA sequence itself. These include:

- [Epigenetic modifications](#), such as DNA methylation.
- Histone modification and [chromatin remodeling](#), which change the accessibility of genes for activation.
- Transcription factors and various forms of RNA, including messenger RNA, microRNA, long non-coding RNA and small interfering RNA.

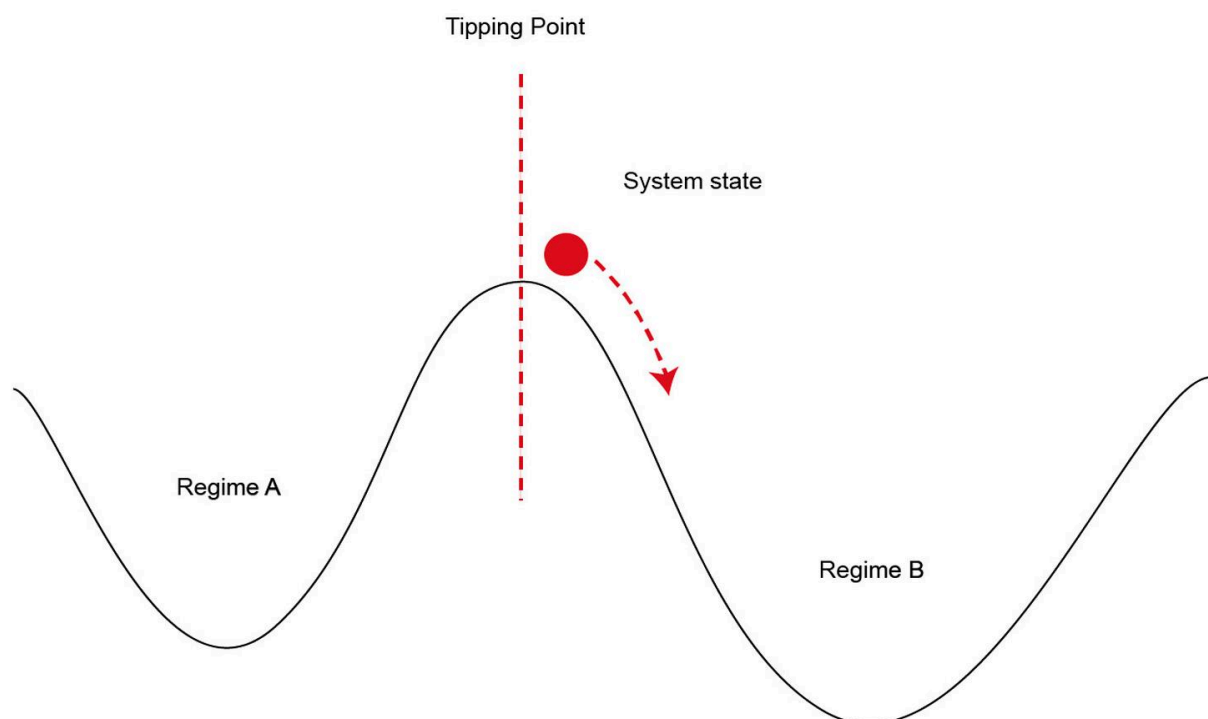
Lung cancer risk factors, as well as random biological events, can alter not only the DNA sequence but also these gene regulating mechanisms. These changes can drive distinct types of malignant transformation, resulting in the diverse histological subtypes observed in lung cancer.

Self-organized criticality

Self-organized criticality describes how complex biological networks maintain relative order while remaining capable of abrupt change. This concept is relevant to malignant transformation in adults and is based on several properties of cellular networks.



[Dropping grains of sand](#) onto a sandpile eventually causes the sandpile to collapse.



[A tipping point](#) is reached when a system tips over from regime A to regime B.

First, cellular networks are [relatively stable](#) because they interact with numerous other networks that constrain rapid change. Due to attractor dynamics, disturbances caused by risk factors often have little immediate effect unless they persist for years to decades. During this time, the system tends to return to a limited set of stable states despite ongoing perturbations.

Second, biological networks may become poised near critical [tipping points](#) that allow rapid shifts in cellular physiology, such as those seen during inflammation, tissue repair or malignant transformation. In a state of [self-organized criticality](#), small disturbances usually have no effect, occasionally produce minor adjustments and only rarely trigger major reconfigurations. This framework helps explain sudden catastrophic events such as earthquakes, [punctuated evolutionary change](#) and malignant transformation. A common analogy is a sandpile onto which grains of sand are dropped one at a time. Most grains cause no visible change, but a single, seemingly trivial grain may trigger a collapse. In this context, rare events rather than steady accumulation can initiate profound biological transitions.

Self-organized criticality contrasts with simple [gradualism](#), in which large scale change is assumed to result from the smooth accumulation of small effects. While gradual processes clearly occur in biology and were supported by [Darwin](#), they [do not fully account](#) for the episodic and nonlinear behavior observed in evolution and cancer development.

Third, during long quiet periods of repeated risk factor exposure, cellular networks may accumulate numerous minor molecular and epigenetic alterations even in the absence of recognizable histologic change. These periods may be followed by [bursts of rapid reorganization](#) that produce premalignant or malignant phenotypes.

From this perspective, cancer emerges as a recurrent consequence of the architecture of complex biological networks in long lived organisms. Although cancer is not inevitable in every individual, its complete elimination at the population level is unlikely. We can reduce its incidence, we can detect it earlier and we can treat it more effectively, but attaining a [world without cancer](#) is not possible.

The superpromoters of cancer

Cancer risk factors appear to act through five major mechanisms that strongly promote malignant transformation. We refer to these mechanisms as [superpromoters](#).

1. [Chronic inflammation](#) destabilizes many biological networks and is difficult to terminate once initiated by persistent risk factors. Cancer risk factors often activate inflammatory pathways without adequately engaging resolution mechanisms. This leads to sustained network activation that may overwhelm redundant control systems and promote malignant transformation through pathways that are difficult to restore to baseline. Chronic inflammation also contributes to carcinogenesis by (a) generating reactive oxygen and nitrogen species that damage DNA, (b) producing [cytokines and growth factors](#) that promote cell proliferation, (c) creating a microenvironment that supports malignant progression and (d) impairing effective immune surveillance.

In lung cancer, chronic inflammation is most commonly related to smoking, occupational carcinogens, air pollutants and tuberculosis. These exposures induce persistent airway inflammation with infiltration by neutrophils and macrophages and increased production of reactive oxygen species. Aging further contributes through a chronic, low grade systemic inflammatory state characterized by elevated circulating proinflammatory cytokines.

2. [DNA alterations and network rewiring](#) modify network structure so that regulatory systems no longer function as intended. These changes may be germline, somatic, random biological events or the result of network rewiring mechanisms such as epigenetic modification or altered mRNA processing. Early in carcinogenesis, these alterations may have minimal phenotypic impact because compensatory controls

remain intact. As additional networks are disrupted and control systems weaken, the same alterations exert progressively greater influence on malignant transformation.

In lung cancer, DNA damage arises from tobacco smoke, occupational exposures, radon and air pollutants including indoor coal combustion, as well as from inherited variants in DNA repair genes. Aging permits the accumulation of mutations from multiple endogenous and exogenous sources.

3. Immune system dysfunction interferes with the elimination of cells that acquire abnormal molecular or phenotypic features. Under normal conditions, immune surveillance identifies and removes many damaged or dysregulated cells. Malignant cells may evade this process by reducing antigen presentation ([camouflage](#)), creating an immunosuppressive microenvironment or disrupting immune signaling pathways.

In lung cancer, smoking and chronic inflammatory exposures appear to be the most important contributors to immune dysfunction. Immune competence also declines with age through [immunosenescence](#). From a [systems biology](#) perspective, the immune system functions as a distributed set of control mechanisms operating throughout [all stages](#) of carcinogenesis.

Public health failures, such as inadequate smoking prevention programs or insufficient access to lung cancer screening and routine medical care, can be viewed as a societal immune system failure. These deficiencies allow many clinically apparent lung cancers to develop that might otherwise have been prevented or detected earlier.

4. Prolonged hormonal exposure is a major promoter of malignant transformation in the breast and prostate but does not appear to play a significant role in lung cancer development.

5. Random biological events affect multiple components of cellular networks. In isolation, these events are usually buffered by redundancy and repair mechanisms. Over years to decades, however, repeated stochastic insults accumulate, producing lasting network damage that alters signaling pathways or shifts cells toward cancer associated attractor states.

Premalignant cancer precursors are not universal

Table 4 - The incidence of malignant precursors varies by cell type and location

Subspecialty	Cancers	Precursors	%
Neuropathology	112	1	0.9%
Bone, joints and soft tissue	140	4	2.9%
Hematopathology	206	6	2.9%
Dermatopathology	79	6	7.6%
Head & neck	128	19	14.8%
Gyn	96	21	21.9%
GI / liver	203	47	23.2%
GU / adrenal	141	34	24.1%
Breast	46	14	30.4%
Thoracic	55	20	36.4%
Grand total	1206	172	14.3%

Totals as of 17Dec2025

Only [14% of the 1,206 types of malignancy](#), based on our definition, have recognized premalignant precursor lesions and their presence varies markedly by cell type and anatomic location (Table 4). Tumors described as undergoing [multistep progression](#) may appear to arise through a linear sequence of increasingly unstable intermediate states. However, the overall process is often nonlinear and the formation of the intermediate states themselves may be discontinuous. For example, many lung adenocarcinomas do not follow a predictable stepwise progression from atypical adenomatous hyperplasia to adenocarcinoma in situ to minimally invasive adenocarcinoma and finally to invasive adenocarcinoma. Instead, alterations in [multiple distinct pathways](#) interact in complex and sometimes unpredictable ways.

In contrast, [colorectal carcinogenesis](#) more commonly follows an apparent linear progression from adenoma to carcinoma, with relatively well characterized and sequential genetic alterations. This difference highlights that precursor frequency and behavior are strongly dependent on tissue context and underlying network structure.

Cancer is an altered systems biology

Advanced cancer in adults may begin with relatively simple molecular changes, but established tumors are sustained by years to decades of supportive network alterations throughout the body. This persistent, body wide reconfiguration can be described as an [altered systems biology](#). Once established, these systemic changes often do not fully revert to normal, even when a substantial proportion of cancer cells are eliminated.

VI. How lung cancer differs from other malignancies:

1. Many lung cancer subtypes (18 of 39 subtypes or 46.2%) have [pre-malignant precursors](#), including [adenocarcinoma](#) and [squamous cell carcinoma](#). In the lung and other [epithelia](#), malignant transformation is slowed by epithelial cell junctions and basement membranes. The epithelial cells may have malignant properties, but they are constrained by epithelial cell - cell interactions ([tight junctions](#), adherens junctions, desmosomes and gap junctions), cell - extracellular interactions and [basement membranes](#) present in epithelial tissue. These interactions limit inappropriate cell division and may force epithelial malignancies to proceed through an [intraepithelial malignancy process](#) before becoming invasive.

Not every malignancy appears to go through the precursor pathway. For example, [atypical adenomatous hyperplasia \(AAH\)](#) is a recognized precursor to both adenocarcinoma in situ and invasive lung adenocarcinoma, but it is identified in only [20% of adenocarcinoma cases](#).

Malignancies without identifiable precursors may develop from clonal expansion of stem or progenitor cells that accumulate genetic and epigenetic alterations, leading to increasing malignant behavior. Over time, as mutations build up, a [tipping point](#) is reached where a critical number of dysfunctional proteins triggers an avalanche of rapid malignant changes, including those disrupting the typical epithelial barriers to malignant spread.

2. Most lung cancer cases are due to cigarette smoking or related risk factors (secondhand smoke or radon acting synergistically with smoking). Targeting cigarette smoking through public health campaigns, regulatory policies and cessation support, together with advances in treatment, has led to dramatic reductions in lung cancer mortality, as noted above.

3. Lung cancers in smokers and nonsmokers are [considered biologically distinct diseases](#). In smokers, tumors develop from the terminal respiratory unit or the central airway compartment. Transformation is driven by *KRAS* mutations in 25% with the remaining cases having no known driver mutations but a higher mutational burden overall and more complex genomic profiles. In contrast, in nonsmokers, tumors develop from the terminal respiratory unit only. Transformation is largely driven by *EGFR* mutations (also *ALK*, *ROS1*, but not *KRAS* mutations). Tumors have a lower overall mutational burden and a mutational signature related to aging and DNA repair.

4. Tumors in nonsmokers, particularly with no other identifiable risk factors, appear to arise from random biological events. This refers to rare, seemingly [random cellular accidents](#) that cause network dysfunction that may propagate to surrounding cellular networks and promote malignancy. These accidents [are more likely](#) after chronic lung injury, inflammation or age related changes and include [DNA replication errors](#) in normal stem cells, [DNA repair defects](#) and [epigenetic alterations](#), mitotic errors affecting [cell component distribution](#) and immune surveillance failure.

5. Lung cancer in smokers often demonstrates a field cancerization effect, which means large areas of histologically normal lung tissue may carry diffuse molecular, genetic and epigenetic changes that predispose it to neoplastic transformation. This explains why multiple, clonally independent foci of atypical adenomatous hyperplasia, adenocarcinoma in situ and invasive adenocarcinoma [can arise within the same lung](#) and why not all precursor lesions are directly related to each other genetically. We speculate that this is due to the diffuse nature of many lung cancer risk factors (smoking, secondhand smoke, radon, air pollution), as opposed to random biological events, which are more likely to be focal. In never smokers, field cancerization is not a typical feature because they lack the intense, diffuse airway exposure to mutagens that produce classical field effects.

VII. How lung cancer arises for all lung cancer types

We have identified [39 distinct types of lung cancer](#).

These malignancies arise through diverse mechanisms, including stepwise neoplastic progression, chromosomal translocations, germline pathogenic variants, gene amplification, malignant transformation of benign lesions, metaplasia and dedifferentiation or transformation from other malignancies. In addition,

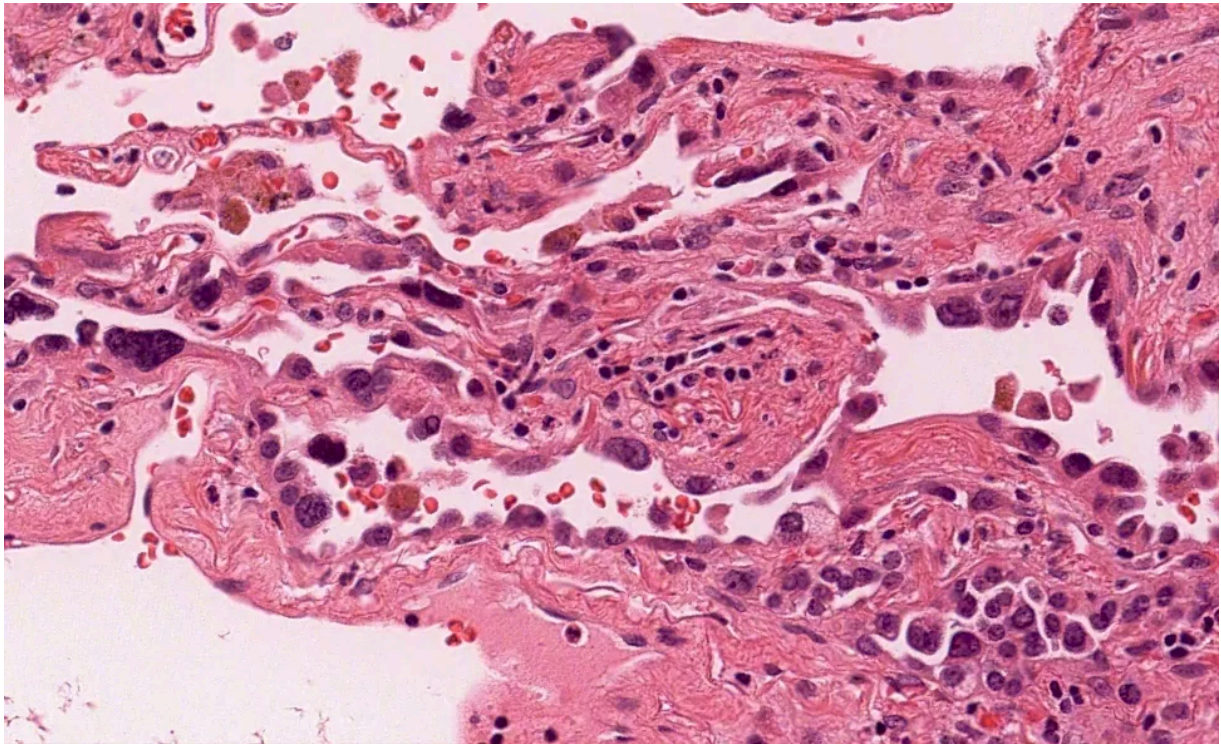
aberrant lineage reprogramming, driven by genetic alterations and microenvironmental influences, can promote divergent differentiation pathways and distinct histologic phenotypes.

Adenocarcinoma

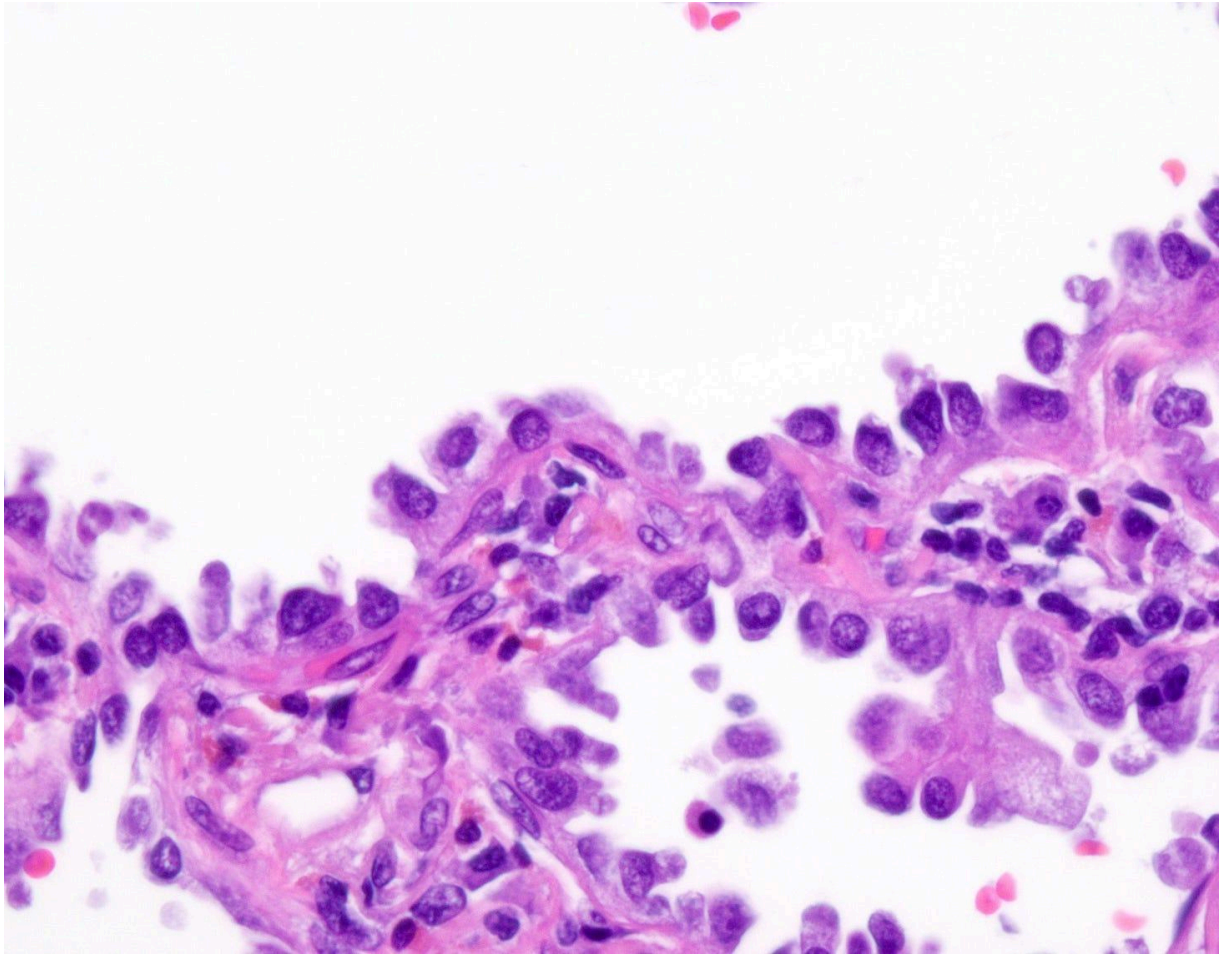


Adenocarcinoma shows gray-white nodules with spicules of black anthracotic pigment. The tumor boundaries may be ill defined grossly due to the lepidic component, which spreads into the surrounding lung parenchyma ([contributed by WebPathology.com](#)).

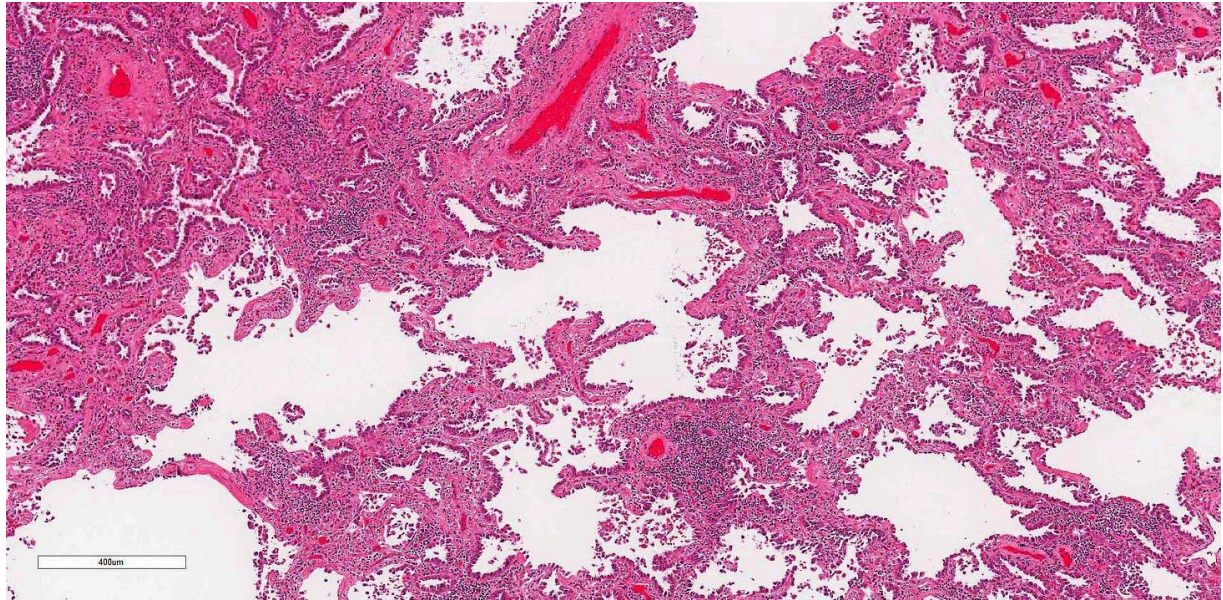
In the United States, adenocarcinoma is the most common type of lung cancer. Nonmucinous adenocarcinoma is thought to frequently follow a classic stepwise progression originating from type II pneumocytes or club cells: atypical adenomatous hyperplasia → adenocarcinoma in situ → minimally invasive adenocarcinoma → invasive adenocarcinoma. In cases without an identifiable precursor lesion, most investigators assume that a precursor existed earlier in tumor evolution but was no longer detectable at diagnosis.



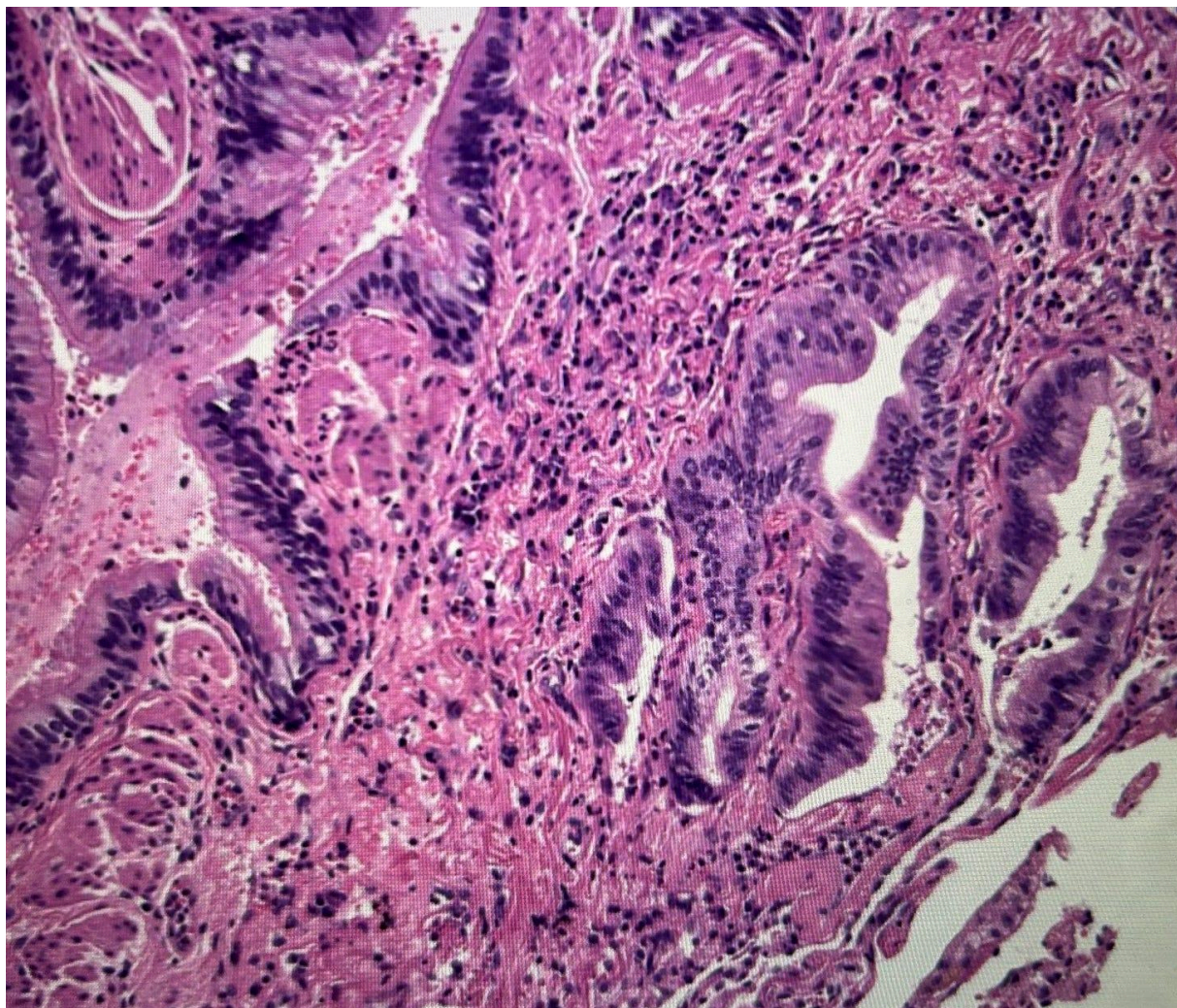
Atypical adenomatous hyperplasia shows a proliferation of atypical pneumocytes that line intact alveolar spaces; no significant architectural distortion or unrelated fibrosis or inflammation is present ([contributed by Sherman Lin and Matthew J. Cecchini, M.D., Ph.D.](#)).



Nonmucinous adenocarcinoma in situ shows back to back continuous lepidic growth of a monolayer of neoplastic cells with focal mild stratification, high nuclear to cytoplasmic ratio, nuclear membrane irregularity, hyperchromasia and anisocytosis; there is no evidence of invasive disease ([contributed by Atreyee Basu, M.D. and Fang Zhou, M.D.](#)).



Nonmucinous minimally invasive adenocarcinoma has a mostly lepidic pattern with irregular glandular structures. There is an intervening desmoplastic stromal reaction and disruption of the normal alveolar architecture ([contributed by Jonathan Keow, M.D., Ph.D.](#)).



Invasive adenocarcinoma, acinar pattern, shows glands invading the stroma. They are irregularly shaped and some are filled with mucin ([contributed by University of Toronto](#)).

Lung adenocarcinoma is frequently associated with [driver mutations](#), defined as recurrent genetic alterations that initiate and sustain malignant behavior. In [one study](#), driver mutations were identified in 90 to 95% of never smokers and 50 to 75% of smokers with lung adenocarcinoma. Commonly affected genes include *EGFR*, *KRAS*, *ALK*, *ROS1*, *BRAF*, *RET*, *MET* and *HER2 (ERBB2)*. These alterations tend to occur early in tumor development and are present in a large proportion of tumor cells, making them effective therapeutic targets. The specific driver mutation often correlates with smoking status, ancestry and histologic features.

There are 12 recognized types of adenocarcinoma in the lung, including 3 “administrative” diagnoses, 5 histologic patterns and 4 rare or uncommon types:

The three administrative diagnoses are primarily practical or classification based and reflect limited tissue sampling or limited invasion rather than a distinct biological subtype.

- [Adenocarcinoma, not otherwise specified \(NOS\)](#), is a diagnosis used for small biopsies or cytologic specimens in which there is insufficient tissue to reliably classify the tumor into established histologic patterns such as acinar, lepidic, papillary, micropapillary or solid.

- [Minimally invasive adenocarcinoma](#), including nonmucinous and mucinous types, is defined as a primary lung adenocarcinoma measuring 3 cm or less with a predominantly lepidic growth pattern and no more than 0.5 cm of stromal invasion. Nonmucinous tumors are commonly associated with *EGFR* mutations and occur more frequently in nonsmokers, women and Asian patients; mucinous tumors more often harbor *KRAS* mutations and are more commonly seen in smokers.

There are five common predominant histologic patterns of lung adenocarcinoma:

- [Adenocarcinoma, acinar predominant](#), is a common histologic pattern characterized by invasive gland-like architecture. It occurs in both smokers and nonsmokers. Many cases show features consistent with stepwise progression from atypical adenomatous hyperplasia through lepidic growth to invasive carcinoma, with accumulation of driver mutations during tumor evolution.
- [Adenocarcinoma, lepidic predominant](#), is strongly associated with never smokers, female sex, East Asian ancestry and *EGFR* mutations. These tumors typically show a lower tobacco related mutational burden. The lepidic pattern reflects persistence of in situ growth along alveolar septa similar to atypical adenomatous hyperplasia and adenocarcinoma in situ. With progression, invasive foci and additional architectural patterns may emerge.
- [Adenocarcinoma, micropapillary pattern](#), is more common in nonsmokers and East Asian patients and is frequently associated with *EGFR* mutations. Micropapillary features are associated with aggressive behavior and poor prognosis. This pattern may arise de novo, through dedifferentiation of acinar or papillary adenocarcinoma or via stepwise progression from lepidic lesions.
- [Adenocarcinoma, papillary predominant](#), is defined by invasive epithelial growth along fibrovascular cores and is not seen in recognized precursor lesions. It is frequently associated with *EGFR* mutations, particularly in nonsmokers. Some tumors show *KEAP1* mutations with activation of the NRF2 pathway. The papillary architecture reflects altered epithelial polarity and tumor stromal interactions.
- [Adenocarcinoma, solid predominant](#), is more commonly associated with smoking and carries a poorer prognosis. These tumors frequently harbor *KRAS* and *TP53* mutations, especially in smokers. *EGFR* mutations are less common in this pattern.

There are four recognized rare or uncommon variants of lung adenocarcinoma:

- [Adenocarcinoma, invasive mucinous type](#), is an uncommon and distinct variant characterized by goblet or columnar tumor cells with abundant intracellular mucin. It likely arises from distal airway or bronchiolar epithelium with mucinous differentiation. *KRAS* mutations are frequent and are thought to promote mucin production and malignant transformation.
- [Colloid adenocarcinoma](#) is a rare variant of invasive lung adenocarcinoma characterized by pools of extracellular mucin containing sparse neoplastic epithelial cells. It may arise in association with mucinous epithelial differentiation of bronchiolar cells. *KRAS* mutations are common in this variant.
- [Adenocarcinoma, enteric type](#), is extremely rare and shows morphologic and immunophenotypic features resembling colorectal adenocarcinoma. Despite this resemblance, these tumors arise in the lung and show molecular profiles that are distinct from typical colorectal carcinomas. The enteric pattern likely reflects aberrant intestinal differentiation acquired during tumor evolution.
- [Adenocarcinoma, fetal type](#), resembles developing fetal lung in the pseudoglandular stage. Low grade and high grade forms are considered distinct entities. Low grade fetal adenocarcinoma is associated with somatic mutations in *CTNNB1* and *DICER1* genes, resulting in Wnt pathway activation and fetal lung-like differentiation. High grade fetal adenocarcinoma occurs predominantly in older men with a history of smoking, is associated with more advanced disease and poorer outcomes and is regarded as a variant of conventional lung adenocarcinoma.

Summary of adenocarcinoma precursors and causes

Malignancy	Precursor	Causes	Environmental or genetic risk factors
Adenocarcinoma, acinar predominant	Atypical adenomatous hyperplasia; adenocarcinoma in situ	<i>EGFR</i> mutations in nonsmokers; <i>KRAS</i> mutations in smokers; stepwise accumulation of driver alterations	Tobacco smoke; environmental carcinogens; female sex; East Asian ancestry in <i>EGFR</i> mutated cases
Adenocarcinoma, lepidic predominant	Atypical adenomatous hyperplasia; adenocarcinoma in situ	<i>EGFR</i> activating mutations; low tobacco associated mutational burden	Never smoking; female sex; East Asian ancestry
Adenocarcinoma, micropapillary pattern	Often none; may evolve from lepidic, acinar or papillary tumors	<i>EGFR</i> mutations; alterations promoting loss of cell polarity and adhesion	Never smoking; East Asian ancestry
Adenocarcinoma, papillary predominant	None recognized	<i>EGFR</i> mutations; occasional <i>KEAP1</i> mutations with <i>NRF2</i> pathway activation	Never smoking; East Asian ancestry
Adenocarcinoma, solid predominant	None recognized	<i>KRAS</i> mutations; <i>TP53</i> mutations; smoking related mutational signatures	Cigarette smoking; occupational inhalants
Adenocarcinoma, invasive mucinous type	None well defined; likely bronchiolar mucinous epithelium	<i>KRAS</i> mutations promoting mucin production and oncogenesis	Smoking in some cases; chronic airway injury
Colloid adenocarcinoma	None well defined	<i>KRAS</i> mutations; mucin related oncogenic pathways	Smoking; chronic airway irritation
Adenocarcinoma, enteric type	None recognized	Aberrant intestinal differentiation programs; variable <i>KRAS</i> or other lung type drivers	Smoking history in reported cases; unknown genetic susceptibility
Adenocarcinoma, fetal type, low grade	None recognized	<i>CTNNB1</i> mutations; <i>DICER1</i> mutations; Wnt pathway activation	Younger age; rare germline <i>DICER1</i> predisposition

Adenocarcinoma,
fetal type, high grade

None recognized

Conventional
adenocarcinoma drivers;
frequent *TP53* alterations

Older age; male sex;
cigarette smoking

Squamous cell carcinoma

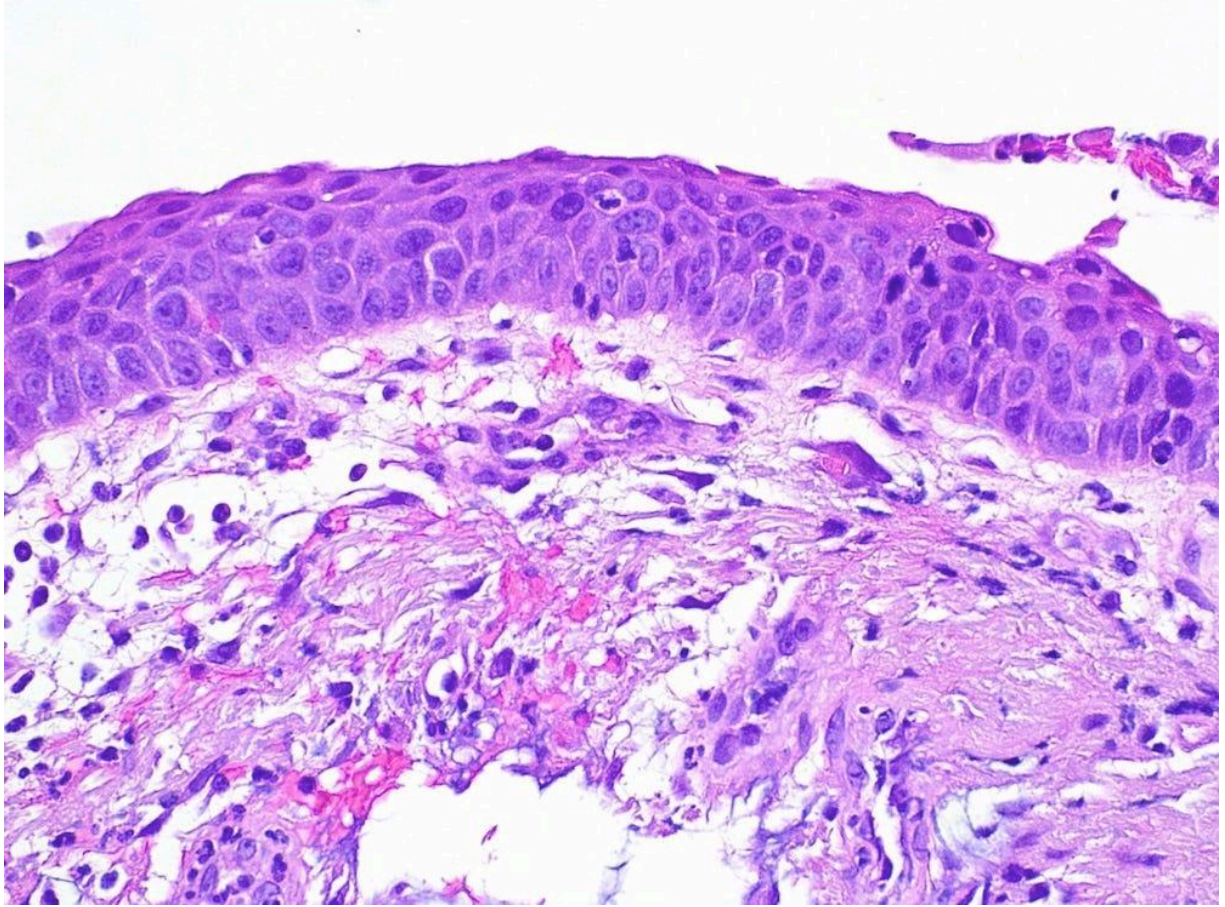


Squamous cell carcinoma, keratinizing, in lower lobe of the left lung ([contributed by @Andrew_Eltv](#))

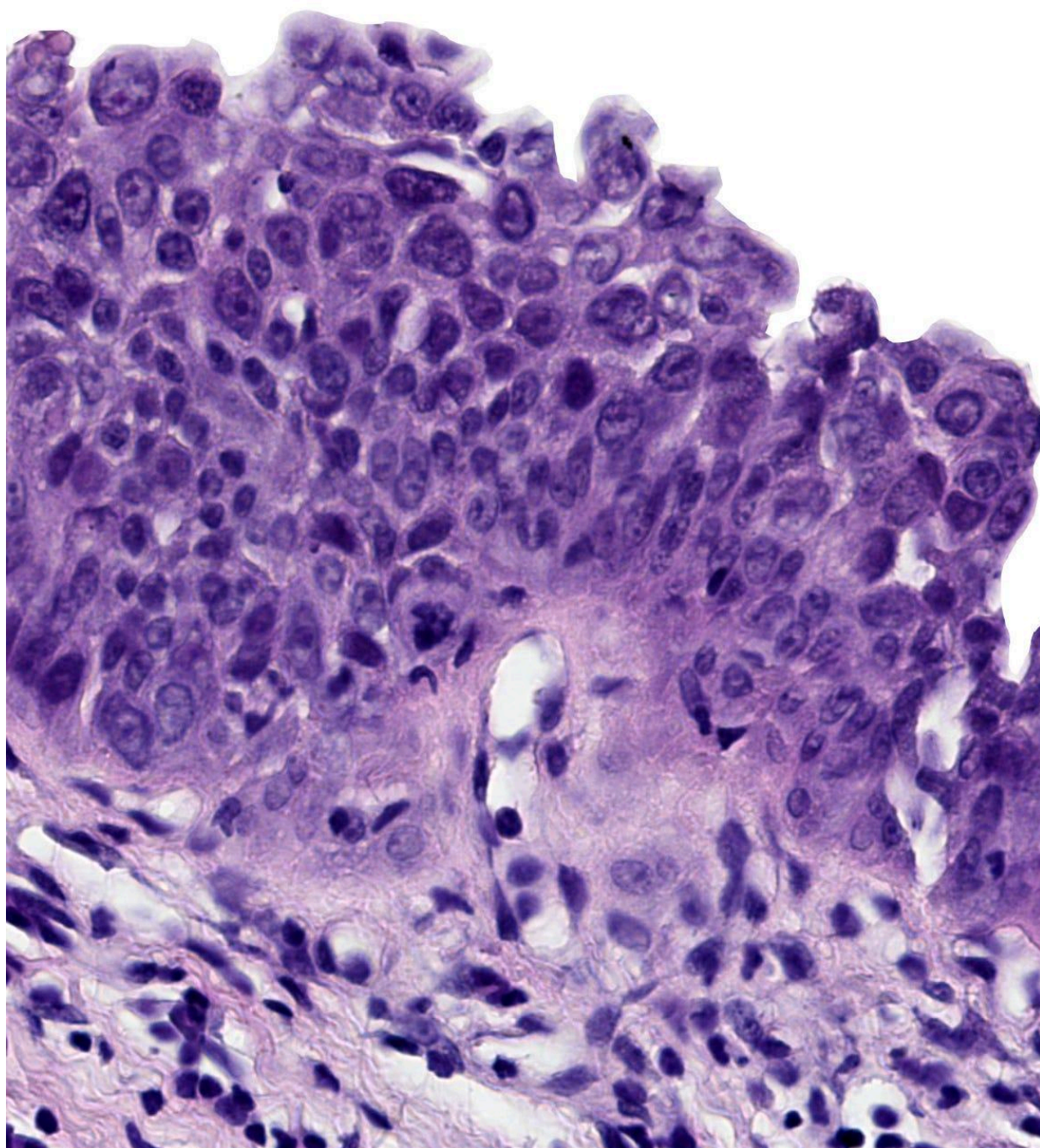
There are 6 recognized types of squamous cell carcinoma in the lung.

[Squamous cell carcinoma](#) is the second most common type of lung cancer in the United States. Its incidence has declined with decreasing tobacco use and it remains more strongly associated with cigarette smoking than adenocarcinoma. Cigarette smoke contains numerous carcinogens that cause widespread DNA damage in basal progenitor cells of the bronchial epithelium, particularly in the central airways. This process creates a field cancerization effect. Tobacco smoke also induces chronic inflammation, oxidative stress and immune dysregulation, which further promote malignant transformation and genomic instability.

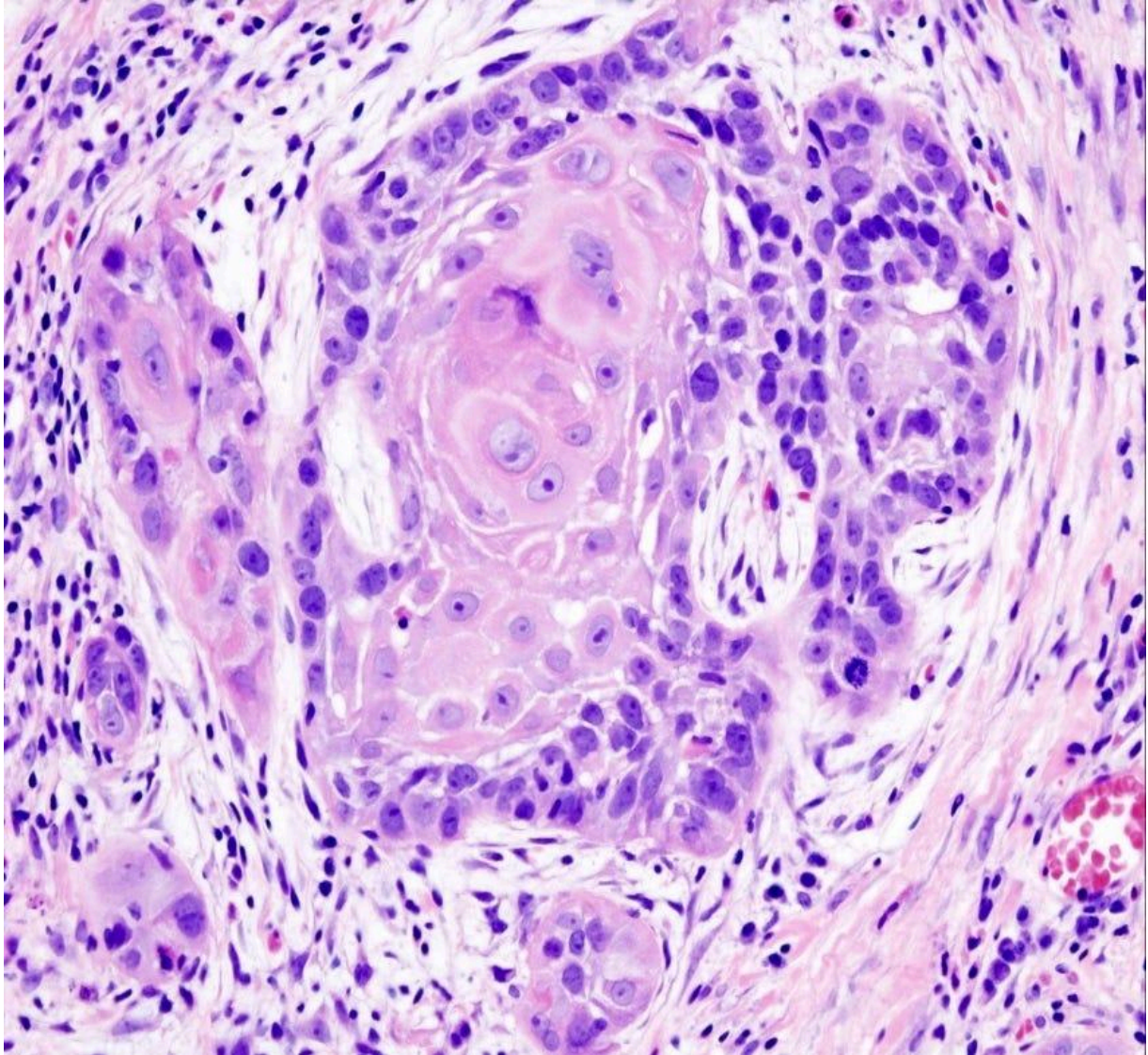
In four common types of squamous cell carcinoma, including squamous cell carcinoma not otherwise specified, keratinizing, nonkeratinizing and basaloid types, tumors typically arise through stepwise progression. Persistent epithelial injury leads to basal cell hyperplasia and squamous metaplasia. With continued genetic damage, squamous metaplasia may progress to squamous dysplasia, squamous cell carcinoma in situ and ultimately invasive squamous cell carcinoma. These tumors often exhibit a high tumor mutational burden, particularly in smokers, resulting in marked intratumoral heterogeneity and the emergence of subclones capable of local invasion, lymph node metastasis and distant spread.



Squamous dysplasia in lung bronchial epithelium ([contributed by Roseann Wu, M.D., M.P.H.](#)).



Squamous cell carcinoma in situ with very pleomorphic (variable) cells ([contributed by Luka Brčić, M.D., Ph.D.](#)).



Squamous cell carcinoma with obvious keratinization ([contributed by Alex McGeough, M.D.](#))

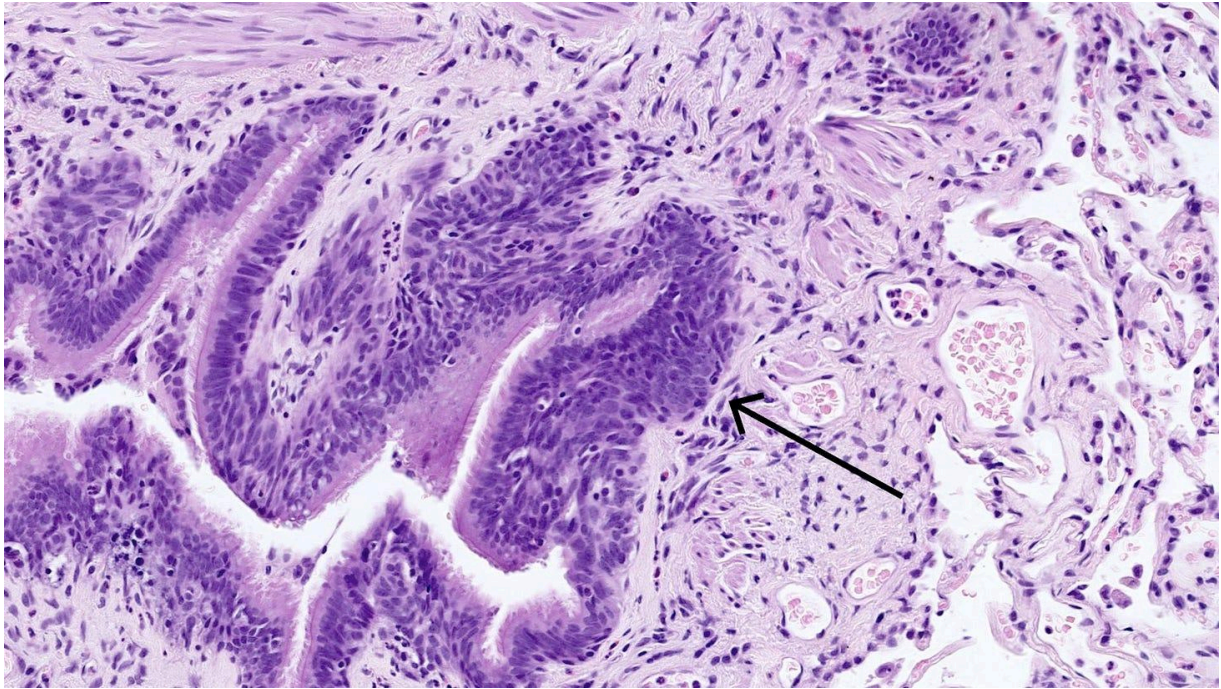
[Lymphoepithelioma-like carcinoma](#) is considered either a rare variant or a distinct subtype of squamous cell carcinoma that resembles [undifferentiated nasopharyngeal carcinoma](#). Its pathogenesis differs by population. In East Asian patients, it more often occurs in younger individuals and nonsmokers and most cases show [Epstein-Barr virus](#) (EBV) positivity in tumor cells by [in situ hybridization](#) or immunohistochemistry. Malignant transformation is thought to occur through EBV related mechanisms including latent infection, immune evasion, clonal expansion and progression to invasive carcinoma. In non Asian populations, this tumor is very rare and is typically not associated with EBV, despite similar histologic features.

[Adenosquamous carcinoma](#) is a rare malignant tumor defined by the presence of at least 10% adenocarcinoma and 10% squamous cell carcinoma components in resection specimens. Molecular and morphologic studies support an origin from a common progenitor cell. It most commonly occurs in male smokers and shares risk factors with other non-small cell lung cancers.

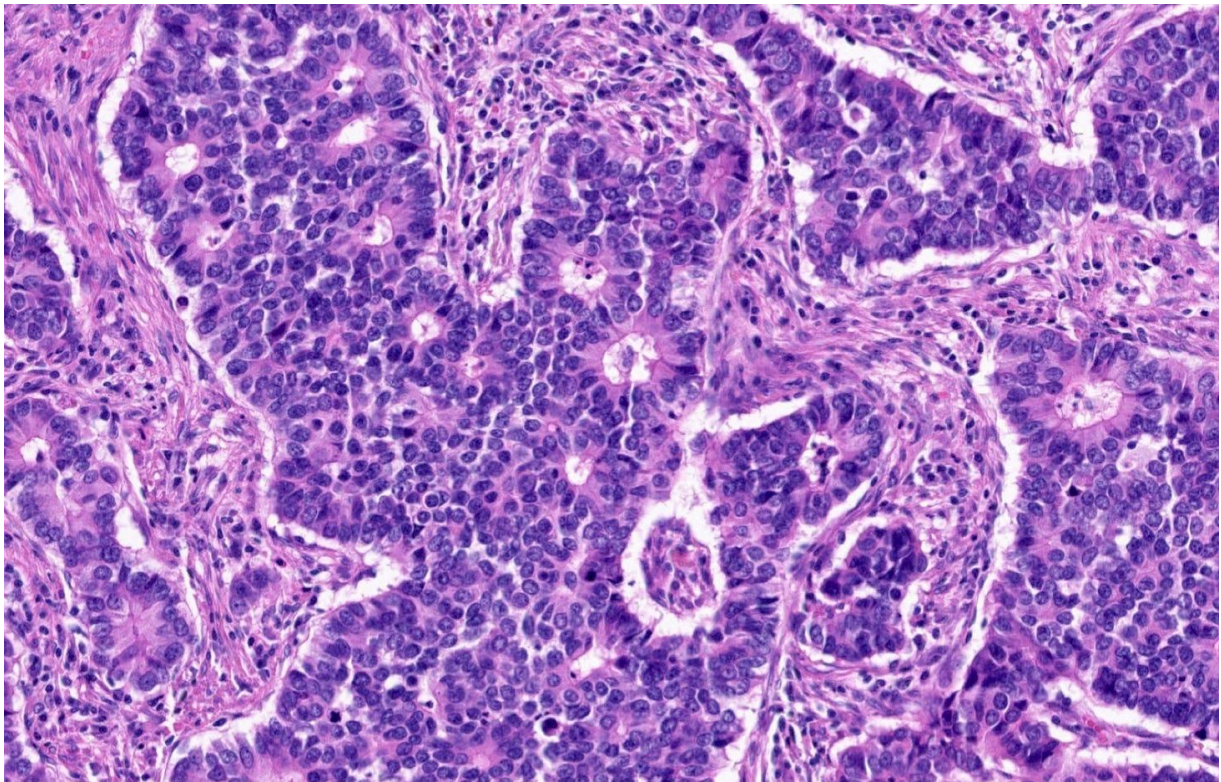
Summary of squamous cell carcinoma precursors and causes

Malignancy	Precursor	Causes	Environmental or Genetic Risk Factors
Squamous cell carcinoma, not otherwise specified	Squamous metaplasia; squamous dysplasia; carcinoma in situ	<i>TP53</i> mutations; <i>CDKN2A</i> loss; <i>SOX2</i> amplification; genomic instability	Cigarette smoking; occupational inhalants; air pollution
Squamous cell carcinoma, keratinizing	Squamous metaplasia; squamous dysplasia; carcinoma in situ	<i>TP53</i> mutations; <i>CDKN2A</i> inactivation; oxidative DNA damage	Heavy tobacco smoking; chronic airway irritation
Squamous cell carcinoma, nonkeratinizing	Squamous metaplasia; squamous dysplasia; carcinoma in situ	<i>TP53</i> mutations; alterations in cell cycle control; occasional <i>PIK3CA</i> mutations	Cigarette smoking; secondhand smoke
Squamous cell carcinoma, basaloid	Squamous dysplasia; carcinoma in situ	<i>TP53</i> mutations; high proliferative signaling; frequent <i>RB</i> pathway disruption	Cigarette smoking; older age
Lymphoepithelioma-like carcinoma	None recognized	EBV driven oncogenesis; activation of immune evasion pathways	East Asian ancestry; Epstein-Barr virus infection; never or light smoking
Adenosquamous carcinoma	Atypical adenomatous hyperplasia for glandular component; squamous dysplasia for squamous component	<i>TP53</i> mutations; <i>KRAS</i> or <i>EGFR</i> mutations in glandular component; divergent differentiation programs	Cigarette smoking; mixed molecular susceptibility

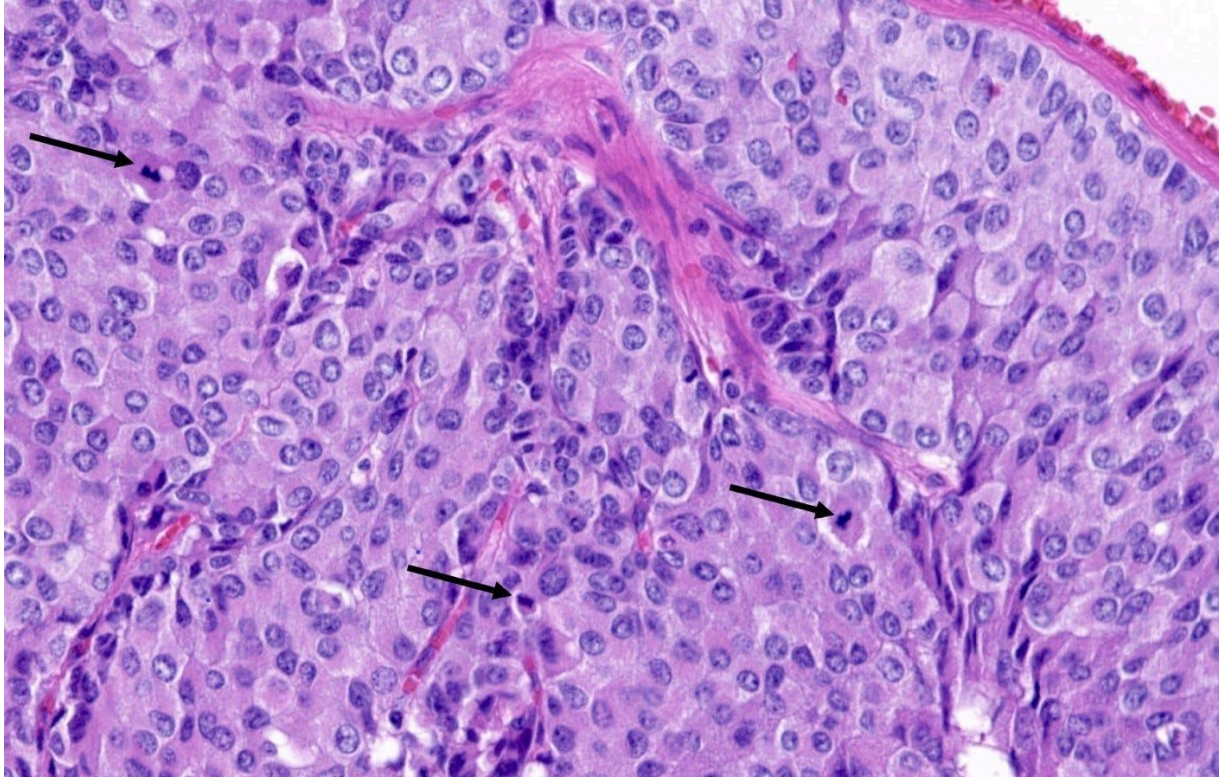
Neuroendocrine tumors



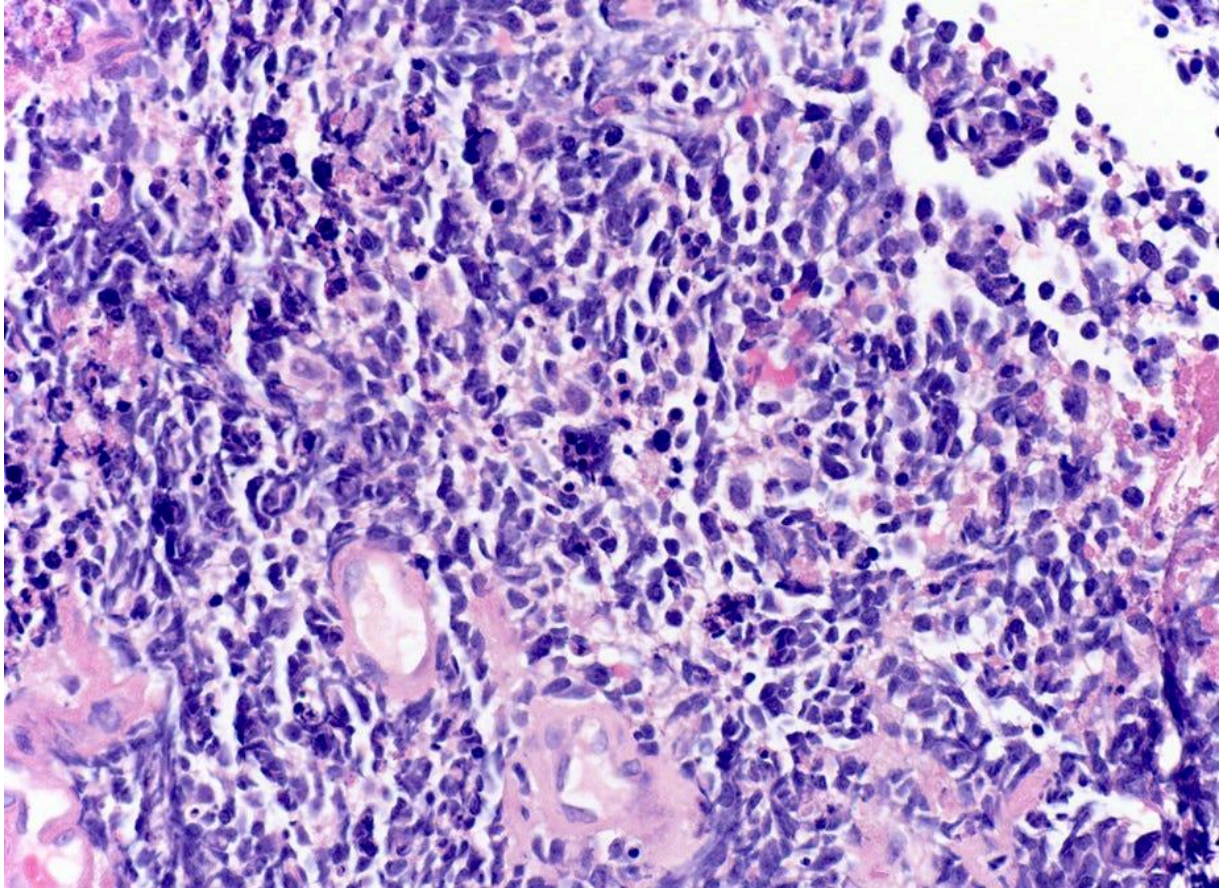
Diffuse idiopathic pulmonary neuroendocrine hyperplasia, a precursor lesion, has proliferating neuroendocrine cells forming small nodules (arrow) limited to the bronchiole ([contributed by Andréanne Gagné, M.D., M.Sc. and Philippe Joubert, M.D., Ph.D.](#)).



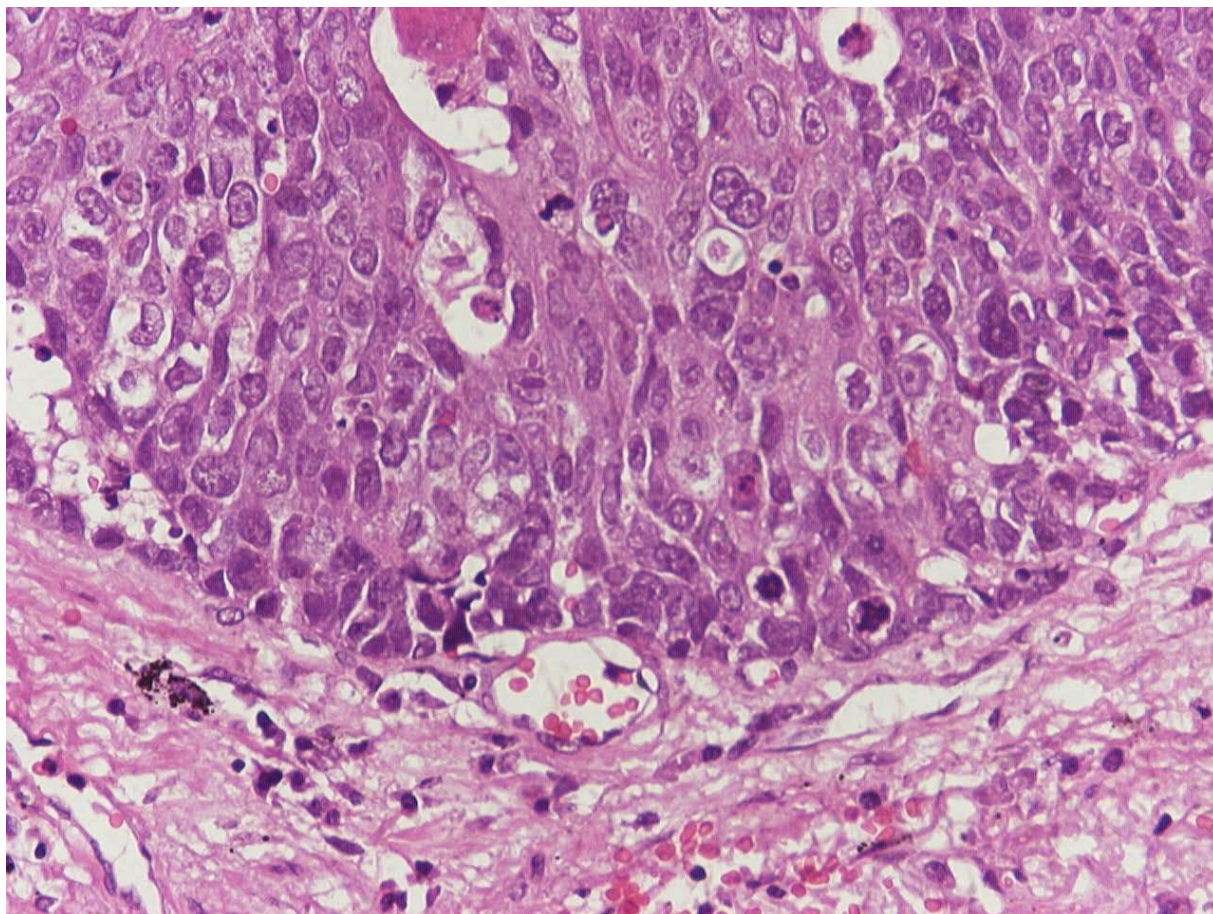
Carcinoid tumor showing rosettes ([contributed by Philippe Joubert, M.D., Ph.D.](#)).



Atypical carcinoid tumor showing 3 mitotic figures ([contributed by Philippe Joubert, M.D., Ph.D.](#)).



Small cell lung carcinoma has crush artifacts and mitotic figures. Tumor cells have finely dispersed or salt and pepper chromatin with no distinct nucleoli ([contributed by Carolyn Glass, M.D.](#))



Large cell neuroendocrine carcinoma has tumor cells with increased nuclear pleomorphism, large nuclei with open chromatin, prominent nucleoli and increased mitotic activity ([contributed by Ioanna Abba Nteka, M.D. and Aggeliki Cheva, M.D., Ph.D.](#)).

There are 4 recognized malignant neuroendocrine tumors in the lung.

[Typical carcinoid tumor](#) is a well differentiated neuroendocrine neoplasm with no established association with cigarette smoking or other environmental exposures. These tumors harbor a low mutational burden and are biologically distinct from small cell carcinoma and large cell neuroendocrine carcinoma. Typical carcinoid tumors arise from pulmonary neuroendocrine cells. Some cases are associated with diffuse idiopathic pulmonary neuroendocrine cell hyperplasia (DIPNECH), a recognized precursor lesion that occurs most often in middle aged to older women without identifiable risk factors. DIPNECH and associated carcinoid tumors may show specific genetic and epigenetic alterations, including involvement of the *MEN1* gene and abnormalities in chromatin remodeling that promote increased proliferation.

[Atypical carcinoid tumor](#) is an uncommon well differentiated neuroendocrine tumor, accounting for 10% of pulmonary carcinoids. It is defined by 2-10 mitoses per 2 mm² or the presence of focal necrosis. Atypical carcinoid tumors have a worse prognosis than typical carcinoid tumors but a significantly better outcome than small cell carcinoma or large cell neuroendocrine carcinoma.

[Small cell carcinoma of the lung](#) is a common high grade neuroendocrine carcinoma that occurs predominantly in smokers and is associated with an aggressive clinical course and poor prognosis. It arises from pulmonary neuroendocrine cells exposed to tobacco related carcinogens and is characterized by near universal inactivation of *TP53* and *RB1*, resulting in profound genomic instability and rapid tumor progression.

[Large cell neuroendocrine carcinoma](#) accounts for 2–3% of all lung cancers. Like small cell carcinoma, it is strongly associated with cigarette smoking and carries a poor prognosis. It is thought to arise from pulmonary neuroendocrine cells damaged by tobacco carcinogens. In rare cases, it may develop through dedifferentiation of a preexisting carcinoid tumor or from a non-small cell lung carcinoma that acquires neuroendocrine differentiation.

Summary of neuroendocrine tumor precursors and causes

Malignancy	Precursor	Causes	Environmental or Genetic Risk Factors
Typical carcinoid tumor	DIPNECH in a subset of cases	Low proliferative drive; altered neuroendocrine cell regulation; epigenetic dysregulation	<i>MEN1</i> alterations; chromatin remodeling abnormalities; female sex; middle aged to older age; no established environmental exposures
Atypical carcinoid tumor	DIPNECH in some cases	Increased proliferative signaling compared with typical carcinoid; focal necrosis	Occasional <i>MEN1</i> alterations; additional cell cycle dysregulation; weak or inconsistent smoking association
Small cell carcinoma	None recognized	Tobacco carcinogen induced genomic damage; loss of cell cycle control	Cigarette smoking; near universal <i>TP53</i> and <i>RB1</i> inactivation; very high mutational burden
Large cell neuroendocrine carcinoma	None recognized; rare dedifferentiation from carcinoid or non-small cell carcinoma	Tobacco related DNA damage; acquisition of neuroendocrine differentiation programs	Cigarette smoking; <i>TP53</i> and <i>RB1</i> alterations; genomic instability

Most of the remaining lung malignancies are rare and have no established risk factors. Many are driven by chromosomal translocations, which are thought to arise as random biological (stochastic) events within a single cell. In general, they occur when double stranded DNA breaks develop in two different chromosomes and the cell's DNA repair machinery erroneously joins mismatched DNA ends. The resulting fusion gene encodes an abnormal protein that can drive malignant transformation.

There are 5 recognized salivary gland-like malignancies in the lung.

These tumors are rare and have no established association with cigarette smoking, environmental exposures or inflammatory lung injury. They arise through malignant transformation of the submucosal seromucinous glands of the large bronchi or trachea, where these glands are most abundant. Many are driven by recurrent chromosomal translocations.

[Adenoid cystic carcinoma](#) of the lung typically harbors the same hallmark gene fusions seen in [salivary gland adenoid cystic carcinoma](#). The most common is the *MYB::NFIB* fusion, usually generated by a

t(6;9) translocation. Less commonly, tumors show a *MYBL1::NFIB* fusion. These alterations act as primary oncogenic drivers.

[Mucoepidermoid carcinoma](#), particularly low grade tumors, is strongly associated with the t(11;19)(q21;p13) translocation resulting in *MAML2* gene rearrangement. This typically involves fusion with *CRTC1* and less commonly *CRTC3*, mirroring [salivary gland counterparts](#). This fusion represents the dominant oncogenic driver in low grade tumors. High grade mucoepidermoid carcinomas may lack *MAML2* rearrangements and show more complex genetic alterations.

[Hyalinizing clear cell carcinoma](#) of the lung is extremely rare, with fewer than 30 reported cases. Most tumors harbor an *EWSR1::ATF1* fusion transcript. In [salivary gland tumors](#), this fusion results from a t(12;22)(q13;q12) translocation. Available evidence suggests that pulmonary tumors share the same underlying mechanism.

[Myoepithelial carcinoma](#) of the lung is extremely rare, with fewer than 60 reported cases. Many myoepithelial carcinomas, particularly those of [salivary gland](#) and [soft tissue](#) origin, harbor *EWSR1* or *FUS* rearrangements with fusion partners such as *PBX1*, *PBX3* or *POU5F1*. Pulmonary cases appear to follow similar molecular patterns, although available data remain limited. Some tumors may arise through malignant transformation of a preexisting [pleomorphic adenoma](#), termed carcinoma ex pleomorphic adenoma.

[Epithelial myoepithelial carcinoma](#) of the lung is extremely rare, with fewer than 100 reported cases. It is a low grade malignancy arising from submucosal bronchial glands and closely resembles its [salivary gland counterpart](#). No consistent or defining genetic alteration has yet been established.

Summary of salivary gland-like malignancy precursors and causes

Malignancy	Precursor	Causes	Environmental or Genetic Risk Factors
Adenoid cystic carcinoma	None recognized	Oncogenic gene fusion driven transcriptional dysregulation	<i>MYB::NFIB</i> fusion from t(6;9); less commonly <i>MYBL1::NFIB</i> ; arising in submucosal seromucinous glands
Mucoepidermoid carcinoma	None recognized	Fusion driven activation of Notch related transcriptional programs in low grade tumors	<i>MAML2</i> rearrangement, most often <i>CRTC1::MAML2</i> from t(11;19); high grade tumors may lack fusion and show complex alterations
Hyalinizing clear cell carcinoma	None recognized	Fusion mediated oncogenic transcriptional activation	<i>EWSR1::ATF1</i> fusion, analogous to salivary gland counterpart; very limited case numbers

Myoepithelial carcinoma	Pleomorphic adenoma in rare cases, carcinoma ex pleomorphic adenoma	Malignant transformation with fusion driven transcriptional dysregulation	<i>EWSR1</i> or <i>FUS</i> rearrangements with partners such as <i>PBX1</i> , <i>PBX3</i> or <i>POU5F1</i> ; limited pulmonary data
Epithelial myoepithelial carcinoma	None recognized	Low grade malignant transformation of submucosal bronchial gland epithelium	No consistent defining genetic alteration identified; molecular pathogenesis incompletely understood

There are 5 recognized mesenchymal malignancies in the lung, all rare.

[Pleuropulmonary blastoma](#) is a malignant tumor of mesenchymal origin that arises in the lung or pleura of children. It is thought to develop from primitive mesenchymal cells that persist abnormally during lung development and undergo neoplastic transformation early in life. The tumor shows embryonal or blastematous morphology. Approximately 65–70% of patients carry a germline *DICER1* mutation; additional cases harbor somatic *DICER1* alterations.

[Pulmonary artery intimal sarcoma](#) is a malignant mesenchymal tumor arising from the intimal layer of large blood vessels, most commonly the pulmonary artery. It is believed to originate from subendothelial mesenchymal cells within the vessel wall. Amplification of *MDM2* is a characteristic molecular finding.

[Primary pulmonary myxoid sarcoma with *EWSR1::CREB1* fusion](#) is a low grade malignant mesenchymal tumor composed of primitive cells showing fibroblastic or myofibroblastic differentiation within a myxoid stroma. It shares molecular and morphologic features with the myxoid variant of [angiomatoid fibrous histiocytoma](#), suggesting a possible biological relationship.

[Malignant PEComa of the lung](#) arises from perivascular epithelioid cells, which have no known normal tissue counterpart and exhibit dual melanocytic and smooth muscle differentiation. Most cases show inactivation of *TSC1* or *TSC2*, either sporadically or in association with tuberous sclerosis complex, resulting in mTOR pathway activation. A subset lacks *TSC1* or *TSC2* alterations and instead harbors *TFE3* gene fusions. Some tumors may arise through malignant transformation of a preexisting benign PEComa. Occasional overlap with [pulmonary lymphangioleiomyomatosis](#) suggests a shared biological origin.

[Epithelioid hemangioendothelioma of the lung](#) is a malignant vascular neoplasm arising from endothelial cells of small vessels. Most cases (90%) are driven by a *WWTR1::CAMTA1* fusion resulting from t(1;3)(p36.3;q25). A smaller subset (10%) harbors a *YAP1::TFE3* fusion.

Summary of mesenchymal malignancy precursors and causes

Malignancy	Precursor	Causes	Environmental or Genetic Risk Factors
Pleuropulmonary blastoma	Abnormal persistence of primitive mesenchyme during lung development	Early life neoplastic transformation of embryonal mesenchymal cells	Germline <i>DICER1</i> mutation in ~65–70% of patients; additional cases with somatic <i>DICER1</i> alterations; childhood onset

Pulmonary artery intimal sarcoma	None recognized	Malignant transformation of subendothelial mesenchymal cells of large vessels	<i>MDM2</i> amplification; location within pulmonary artery or major branches
Primary pulmonary myxoid sarcoma with <i>EWSR1::CREB1</i> fusion	None recognized	Fusion driven oncogenic transcriptional activation	<i>EWSR1::CREB1</i> fusion; biologic overlap with myxoid angiomatoid fibrous histiocytoma
Malignant PEComa	Benign PEComa in some cases	mTOR pathway activation in perivascular epithelioid cells	Inactivation of <i>TSC1</i> or <i>TSC2</i> sporadically or with tuberous sclerosis complex; subset with <i>TFE3</i> gene fusions; overlap with lymphangioleiomyomatosis
Epithelioid hemangioendothelioma	None recognized	Fusion driven dysregulation of Hippo signaling and endothelial differentiation	<i>WWTR1::CAMTA1</i> fusion in ~90% of cases; <i>YAP1::TFE3</i> fusion in a minority

There are 3 recognized sarcomatoid carcinomas of the lung, all rare.

[Pleomorphic carcinoma](#) is a high grade non-small cell carcinoma containing both conventional epithelial and sarcomatoid components. It is thought to arise from a preexisting non-small cell lung carcinoma through epithelial-mesenchymal transition, which leads to the development of spindle or giant cell elements.

[Pulmonary blastoma](#) is an uncommon biphasic tumor, primarily seen in children and young adults. It is hypothesized to originate from primitive pulmonary epithelial or multipotent progenitor cells, often driven by recurrent mutations in *CTNNB1* (β -catenin).

[Carcinosarcoma](#) is a malignant tumor containing both epithelial and mesenchymal elements. It is believed to arise from a common clonal epithelial precursor that undergoes epithelial-mesenchymal transition. *TP53* mutations are frequently observed.

Summary of sarcomatoid carcinoma precursors and causes

Malignancy	Precursor	Causes	Environmental or Genetic Risk Factors
Pleomorphic carcinoma	Preexisting non-small cell lung carcinoma	Epithelial-mesenchymal transition with acquisition of spindle and giant cell morphology	Cigarette smoking; genomic instability; frequent <i>TP53</i> alterations; complex mutational profiles

Pulmonary blastoma	None clearly established; hypothesized primitive pulmonary epithelial or multipotent progenitor cells	Aberrant embryonal signaling with biphasic epithelial and mesenchymal differentiation	Recurrent activating mutations in <i>CTNNB1</i> (β -catenin); childhood and young adult onset
Carcinosarcoma	Preexisting epithelial carcinoma	Clonal epithelial tumor undergoing epithelial-mesenchymal transition with mesenchymal differentiation	Cigarette smoking; frequent <i>TP53</i> mutations; marked genomic instability

There are 4 additional recognized lung malignancies.

[Large cell carcinoma](#) is an uncommon high grade non-small cell carcinoma, accounting for 8% of lung cancers in men and 6% in women, although the reported incidence is declining. It is a diagnosis of exclusion, made on resected specimens that show no morphologic or immunohistochemical evidence of glandular, squamous or neuroendocrine differentiation. Molecular studies show that some tumors share mutations with poorly differentiated adenocarcinomas or squamous carcinomas, suggesting they may represent dedifferentiated forms.

[Primary melanoma of the lung](#) is exceedingly rare, with fewer than 50 reported cases. Metastatic melanoma to the lung is far more common. Some authors have suggested that many tumors reported as primary may actually represent metastases from regressed cutaneous or mucosal primaries, based on mutational signatures and clinical findings.

[Langerhans cell histiocytosis](#) is a rare clonal proliferation of dendritic Langerhans cells. It is frequently linked to cigarette smoking, which induces airway injury, recruits dendritic cell precursors and generates a cytokine-rich microenvironment that can promote oxidative DNA damage. A subset of cells acquires somatic activating mutations in the MAP kinase pathway, including *BRAF* (*V600E*), *KRAS*, *NRAS* and *MAP2K1*.

[NUT \(NUTM1\) carcinoma](#) is a rare, aggressive malignancy that typically occurs in adolescents and young adults. Most cases are driven by a t(15;19)(q14;p13.1) translocation producing a *BRD4::NUTM1* fusion gene, although other fusion partners have been described.

Summary of additional lung malignancy precursors and causes

Malignancy	Precursor	Causes	Environmental or Genetic Risk Factors
Large cell carcinoma	None recognized; may represent dedifferentiation of non-small cell lung carcinoma	Loss of lineage specific differentiation programs in high grade carcinoma	Cigarette smoking; genomic instability; mutations overlapping with poorly differentiated adenocarcinoma or squamous carcinoma

Primary melanoma of the lung	None established; many reported cases likely metastatic from occult or regressed primary	Malignant transformation of melanocytic cells of uncertain origin	Possible shared mutational signatures with cutaneous or mucosal melanoma; extreme rarity argues against a distinct etiologic pathway
Langerhans cell histiocytosis	Cigarette smoke induced airway injury with dendritic cell recruitment	Clonal expansion of Langerhans cells with MAP kinase pathway activation	Cigarette smoking; activating mutations in <i>BRAF</i> (V600E), <i>KRAS</i> , <i>NRAS</i> , or <i>MAP2K1</i>
NUT carcinoma	None recognized	Fusion driven epigenetic dysregulation blocking differentiation	<i>NUTM1</i> rearrangement, most often <i>BRD4::NUTM1</i> from t(15;19); adolescents and young adults

VIII. Treatment strategies to substantially reduce lung cancer deaths

To meet our [strategic goal](#) of reducing U.S. cancer deaths from 600,000 to 100,000 annually, deaths must drop by 83%. For lung cancer, this would mean a reduction from a projected 124,730 deaths in 2025 to roughly 21,200. Most of this reduction could theoretically be achieved by eliminating tobacco smoking and secondhand smoke, which account for 75% and 5% of cases, respectively. Other related risk factors, including indoor air pollution, chronic obstructive pulmonary disease and the synergistic effects of smoking with radon, outdoor air pollution, asbestos and workplace carcinogens, would also contribute. However, risk declines gradually after smoking cessation, with an elevated risk persisting for up to [15 years](#).

In addition, understanding lung cancer's origins, genetic, molecular and environmental, is essential for developing effective treatments for all patients, regardless of smoking history.

Our strategic plan to substantially reduce cancer deaths, including lung cancer deaths, is essentially a management plan to use all of the possible treatment and public health approaches that may impact these deaths. Although targeted therapies and immune checkpoint inhibitors have achieved major successes, no single “silver bullet” is likely to eliminate most adult cancer cells or restore order across the many networks disrupted in malignancy. Instead, success will likely arise from an expanding number of modest improvements with occasional bursts of major advances.

We recommend the following strategies:

1. Target multiple networks driving malignancy:

Curative treatment must address both tumor intrinsic and microenvironmental factors, including:

- Cell cycle regulation, growth factor signaling, DNA repair, apoptosis and related pathways
- The tumor supportive microenvironment at primary and metastatic sites, including vasculature, inflammatory cells, mediators and extracellular matrix
- Microenvironments that maintain cancer cells in embryonic or dedifferentiated states
- Chronic inflammation linked to cancer risk factors
- Immune dysfunction
- Germline genetic alterations that facilitate tumor progression
- Other biological systems that sustain malignancy
- [Synthetic lethality](#), in which simultaneous mutations in two genes lead to cell death

- Strategies that shift tumor cell networks into [less aggressive states](#), see also [here](#).

Because cancer operates through redundant pathways, multiple agents are often required for each therapeutic target. Single agent therapy is frequently bypassed, while combinations can destabilize malignant networks. Examples include:

- [Pembrolizumab plus chemotherapy](#) for PDL1 negative non-small cell lung cancer
- [Platinum doublet chemotherapy](#), where each agent alone has limited efficacy
- [VEGF inhibitors](#) combined with EGFR tyrosine kinase inhibitors.

[We speculate](#) that effective regimens for each adult cancer type exist, even for aggressive tumors, but may require 10-15 therapies that are partially effective individually but synergistic together. Developing 30 or more partially effective therapies and using machine learning, experimental models and clinical trials will help evaluate the many possible combinations.

2. Long term success requires minimizing exposures that promote premalignant and malignant conditions, as discussed in [The American Code Against Cancer](#) or similar guidelines. For lung cancer, this includes reducing tobacco use, air pollution and occupational carcinogen exposure.
3. We must better identify premalignant or malignant lesions through more effective screening, including improvements in low dose CT (LDCT) screening, molecular and liquid biopsy screening, detecting premalignant changes in bronchial epithelium and possibly AI based screening algorithms.
4. We should enact public policy that [strengthens public health programs](#) and optimizes access to cancer prevention, screening and treatment. We must reduce disparities related to race, ethnicity, geographic region and socioeconomic status.
5. Patients should be advised and taught to avoid [medical misinformation](#), to [be organized regarding medical issues](#) and to use sound medical judgment.
6. We should expand clinical trials to determine the effectiveness of individual therapies and combinations, how best to incorporate the large number of combinatorial therapies into a therapeutic regimen, how to reduce treatment side effects and what adjustments need to be made for particular patients. The history of curative cancer therapy in children and young adults suggests that protocols originally thought to be too difficult to implement can be simplified and made more tolerable. Toward this end, every patient should be considered for a clinical trial.